Price distortions and hoarding: An experiment*

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March 31, 2026

Abstract

We study the implications of current and anticipated price distortions in a two-period model in which individuals can store goods. Both expected rationing and expected liberalization generate incentives to hoard. We test these predictions in the laboratory using three treatments that vary whether and when a price cap is imposed: a free market with rising costs, an anticipated price cap, and a liberalization scenario. The experiment yields three main findings. First, the model accurately predicts patterns of shortages and prices across treatments. Second, when price controls are anticipated, participants initially store less than predicted and bid too little for storage, as if they underestimate the severity of future rationing, though storage behavior improves with experience. Third, the free market achieves the highest total surplus, while both price-control treatments generate substantial welfare losses: the liberalization scenario, where rent-seeking rather than defensive storage predominates, produces the lowest total surplus of the three treatments.

Keywords: price controls, price distortions, hoarding, lab experiments.

JEL Classification: D41, C92

*We would like to thank Mikhail Freer, Shan Gui, Roulong Xiao and participants at several audiences for their comments.

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1 Introduction

Where there are price controls, there is hoarding. In Venezuela, years of price controls and shortages led families to stockpile goods at home, even as politicians declared war on hoarders.[1] In Argentina, government-imposed price controls aimed at curbing inflation produced the same response (National Public Radio, 2023). But hoarding is not confined to inflation-plagued economies or to the defunct Soviet Union. As Guenette (2020) documents, price controls are more prevalent than commonly believed, affecting both developed and emerging economies. With the onset of the COVID-19 pandemic, most countries experienced a similar pattern of shortages and hoarding, and in the United States officials invoked anti-price gouging laws against resellers (Rydzewski, 2020; Department of Justice, 2022).

Weitzman (1991) explained the perpetual scarcity of goods in the Soviet Union, where items always seemed in short supply yet somehow accessible: when markets cannot reach equilibrium prices, the effort to secure goods becomes excessively high, and rational individuals respond by stockpiling. The same logic applies to hoarding during the pandemic: the high cost of visiting markets while an infectious disease spreads prompts people to reduce shopping frequency, and sluggish price adjustments by sellers contribute to shortages. More broadly, whenever rationing announcements signal future scarcity, they can trigger the very hoarding they aim to prevent.[2]

In this paper, we model hoarding as a rational response to price distortions and test our predictions experimentally. The key mechanism is the uncertainty generated by rationing: when a price cap holds the market price below equilibrium, demand exceeds supply, and each buyer faces a positive probability of being unable to purchase. This rationing risk creates an incentive to store goods, which in turn affects equilibrium prices, quantities, and efficiency.

[1] See e.g. Wall Street Journal (2016). Figure 14 in the appendix shows a picture of a Caracas supermarket with a sign that reads "no more than 6 cans of tuna per person, thank you" in 2014.

[2] A vivid example occurred in Bogota during the 1997–98 El Niño drought. When authorities implemented a water rationing program to protect reservoir levels, each rationing announcement prompted Bogotans to rush and fill improvised water tanks at home, paradoxically increasing consumption. The crisis was ultimately resolved not by rationing but by raising water utility prices (Serrato et al., 2011; Ivanova and Sarmiento, 2013). In Colombia, water supply is a regulated monopoly; the Water Regulatory Commission (CRA) modified its pricing methodology through Resolution 15 of 1997 to allow price increases in larger cities. Over the following 12 years, average water consumption fell by 17%.

We propose a two-period model with storage and study three scenarios: a free market with rising costs, a market that anticipates future price controls, and a market that anticipates liberalization. We validate the predictions with laboratory experiments.

Prices serve as mechanisms that facilitate cooperation and convey information among economic actors (Hayek, 1945). When price controls are imposed, they disrupt these mechanisms, leading to resource misallocation and coordination problems (Weitzman, 1974; Yohe, 1978). While some market anomalies may justify regulation, hastily implemented price controls often worsen existing issues, encouraging rent-seeking behaviors and non-price competition (Wedig et al., 1989; Deacon and Sonstelie, 1989; Hatfield et al., 2016; Brekke et al., 2006; Coyne and Coyne, 2015; Guenette, 2020). Laboratory experiments suggest that inefficiencies arising from price controls surpass predictions from standard partial equilibrium models, potentially displacing high-surplus consumers. Even when price controls are not explicitly restrictive, they can distort the price discovery process, affecting consumer welfare (Smith and Williams, 1981; Isaac and Plott, 1981; Coursey and Smith, 1983).

Our model predicts that, in a free storage market, the sum of the initial price and storage costs equals the price in the second period. When price distortions emerge, we term the equilibrium the "rationing equilibrium" due to the optimal level of rationing that occurs, even when hoarding goods is viable. In this equilibrium, high-reservation buyers store goods during the first period, while low-reservation buyers purchase at the second period's price ceiling. The combined costs in the first period exceed this price ceiling. Furthermore, the first-period price cannot be lower than the competitive price, and the quantity stored under rationing is equal to or exceeds that in a competitive framework.

The case of market liberalization depends on the rationing rule adopted. We selected a rationing rule that assigns the same probability to the demand for each good to be effectively satisfied. This encourages buyers to try to buy in excess to secure lower prices and maximize the chances of securing their current consumption causing shortages.[3] Due to rationing, only

[3]This is exactly what happened in Argentina when President Carlos Menem announced a 25% increase in gasoline prices due to liberalization in 1990. This announcement triggered long queues at service stations and gasoline shortages throughout the country (Archivo DiFilm, 1991).

a fraction of these attempts result in effective hoarding. However, this is sufficient to drive the future price lower than in the static or *myopic* equilibrium.

To validate our core findings, we conducted experiments in three treatments. The supply shock treatment implements a free market where rising production costs create incentives for storage but not rationing. The SS + Price Cap treatment starts with a free market and introduces a price cap in the second period, encouraging defensive storage. The liberalization treatment imposes a price cap in the first period followed by a free market, generating shortages and rent-seeking incentives.

Our experiments revealed that participants stored fewer units than predicted in treatments with price controls, yet bid too little for them. In the SS + Price Cap treatment, participants stored 2.4 units per market instead of the predicted 3, and the first-period price averaged 74 instead of the predicted [88, 90], as if they underestimated the tightness of the future rationed market and expected to access it anyway. Trading prices generally align with the expected range: in the supply shock treatment, prices in both periods fall within the competitive equilibrium; in the SS + Price Cap treatment, the first-period price falls below the predicted range while the second-period price matches the cap; and in the liberalization treatment, the price cap binds in the first period and the second-period price more than doubles to 109 once the market is freed.

We also accurately predicted the patterns of shortage. We did not observe meaningful shortages in the supply shock treatment. In the SS + Price Cap treatment, the second period (where the cap binds) shows a 60% shortage rate. In the liberalization treatment, the first period shows a 25% average shortage rate, which worsens over rounds as participants learn to exploit the regulated price. Crucially, periods without a price cap show near-zero shortage rates across all treatments. Measured in terms of total surplus (the sum of buyer and seller surplus, net of storage costs), the free market generates the highest welfare (701 LCU per market), followed by the SS + Price Cap treatment (652 LCU) and liberalization (609 LCU). All three treatments achieve similar efficiency ratios relative to their treatment-specific theoretical benchmarks (77%, 75%, and 77%, respectively), but the absolute welfare

losses under price controls are substantial. The liberalization treatment produces the lowest total surplus, driven by the combination of first-period rationing, rent-seeking storage, and a sharp post-liberalization price spike.

1.1 Related literature

The phenomenon of hoarding has been studied in the past. Arrow et al. (1951) explored optimal inventory policies in a seminal work, laying the foundation for understanding storage costs' role in market dynamics. Although the theoretical literature on market equilibrium under price controls is abundant, most works focus on static scenarios, such as Drèze (1975), Madden (1984), and Nguyen and Whalley (1986). Recently, Hatfield et al. (2012) and Herings (2018) use matching models to study competitive equilibrium with price controls.

Our work is most related to Weitzman (1991). We differ from Weitzman in the mechanism that drives hoarding. While Weitzman's model relies on transaction costs, our model is based on the probability of shortages as the primary incentive for hoarding goods. Both explanations align with observations during crises. As mentioned earlier, shortages and transaction costs can account for hoarding behavior during the COVID19 pandemic. Hansman et al. (2020) provides some empirical evidence that sticky prices encourage the hoarding of goods by households, even in the absence of traditional panic or precautionary motives for hoarding.

To the best of our knowledge, our model is the first one to combine the concepts of hoarding and price controls in a simple two-period setting, ideal for a laboratory implementation. Our model is broadly applicable to price distortions, and versions of it can be applied to the study of monetary economics in a similar way to the approach suggested by Clower (1967) and Kemp and Long (1990).

The rest of the paper is organized as follows. In Section 2, we offer a comprehensive description of the model and present two theorems summarizing the theoretical findings. The experimental design and hypotheses are detailed in Section 3. Section 4 discusses the results of the experiment. And Section 5 concludes. For supplementary graphs, experimental

instructions, and related materials, please refer to the online Appendix.

2 Conceptual framework

2.1 An economy with storage

We propose a simple model to explore the effects of price controls when storage is available. We consider an economy with two goods, "money" and a traded good that is available only in indivisible units. There is a large number of traders (a continuum), given by the interval $[0, 1]$. We let $B = [0, n_B]$ and $S = [n_B, 1]$ be the set of buyers and sellers, respectively, so that the fraction of the population that are buyers and sellers are equal to $0 < n_B < 1$ and $n_S = 1 - n_B$, respectively. All traders live for two periods $t = 1, 2$.

The utility function of each buyer $i \in B$ is

$$u_i(q_{i1}, q_{i2}, m_i) = r_i(q_{i1} + q_{i2}) + m_i,$$

where $r_i > 0$ is a parameter representing the buyer's reservation value for the traded good, $q_{it} \in \{0, 1\}$ is the buyer's consumption of the traded good in period t, and $m_i \in \mathbb{R}$ is the buyer's (final) money holding. Buyers enjoy consuming at most one unit of the traded good per period, but they can store one unit in period 1 for consumption in period 2 at a (monetary) cost of $s > 0$. Note that we allow for negative money holdings; we normalize the endowment of money of all individuals to zero. We assume that buyers are strictly ordered by their reservations; in particular, $r_i = r(i)$, where $r(\cdot) : [0, n_B] \to \Re_+$ is a continuous and strictly decreasing function.

Similarly, the utility function of each seller $j \in S$ is

$$u_i(y_{j1}, y_{j2}, m_j) = -c_{j1}y_{j1} - c_{j2}y_{j2} + m_j,$$

where $c_{jt} > 0$ is a parameter representing the seller's cost of production in period t, $y_{jt} \in \{0, 1\}$ is the seller's production of the traded good in period t, and $m_j \in \mathbb{R}$ is the seller's

(final) money holding. Note that each seller's production in each period is capped at one. We allow for changes in the cost of production so as to represent supply shocks. We assume that sellers are strictly ordered by their costs; in particular, $c_{jt} = c_t(j)$, where $c_t(\cdot) : [n_B, 1] \to \Re_+$ is a continuous and strictly increasing function for $t = 1, 2$. To avoid dealing with several corner cases, we assume

$$r_0 > \max(c_{n_B,1}, c_{n_B,2}) \quad \text{and} \quad r_m < \min(c_{n_B+m,1}, c_{n_B+m,2}), \text{ where } m = \min(n_B, n_S). \quad \text{(A)}$$

2.2 Competitive equilibrium

We first consider a situation in which individuals may trade freely at market prices. We let x_{i1} and x_{i2} be the quantity of the traded good acquired by buyer i in period 1 and in period 2. Since a buyer can consume at most one unit in period 1 but can store one unit for later consumption, choosing $x_{i1} = 2$ allows the individual to consume $q_{i1} = q_{i2} = 1$ at a monetary cost of $2p_1 + s$. Given prices $p_1 \geq 0$ and $p_2 \geq 0$ in periods 1 and 2, respectively, the problem of each buyer $i \in B$ is then

$$\max_{x_{i1}\in\{0,1,2\}, x_{i2}\in\{0,1\}} r_i(x_{i1} + x_{i2}) - p_1 x_{i1} - p_2 x_{i2} - s \max\{x_{i1} - 1, 0\},$$

and the problem of each seller $j \in S$ is

$$\max_{y_{j1}\in\{0,1\}, y_{j2}\in\{0,1\}} (p_1 - c_{j1}) y_{j1} + (p_2 - c_{j2}) y_{j2}.$$

Market clearing conditions are given by

$$\int_{i\in B} x_{i1} di = \int_{j\in S} y_{j1} dj \quad \text{and} \quad \int_{i\in B} x_{i2} di = \int_{j\in S} y_{j2} dj.$$

A <u>competitive equilibrium</u> is a demand vector (x_{i1}, x_{i2}) for each buyer $i \in B$, a supply vector (y_{j1}, y_{j2}) for each seller $j \in S$, and a price vector (p_1, p_2) such that buyers' demand and sellers' supply solve their respective problems, and markets clear.

To find the competitive equilibrium, we start by analyzing each market separately. As usual, we let the supply and demand functions in each period $t = 1, 2$ be given by

$$S_t(p) \equiv c_t^{-1}(p) \text{ and } D_t(p) \equiv r^{-1}(p).$$

We refer to prices p_1^0 and p_2^0 such that $S_1(p_1^0) = D(p_1^0)$ and $S_2(p_2^0) = D(p_2^0)$ as "myopic" since they occur when buyers ignore the possibility of storage. Under assumption (A), myopic prices are unique and $p_1^0, p_2^0 \in (r_m, r_0)$.

If $p_1^0 + s \geq p_2^0$, then myopic prices are competitive, since buyers would not gain by storing in period 1. If $p_1^0 + s < p_2^0$, we can find the competitive equilibrium by considering a market for stored goods as follows. Let

$$\tilde{S}(p_1) = S_1(p_1) - D(p_1) \quad \text{and} \quad \tilde{D}(p_1) = D(p_1 + s) - S_2(p_1 + s)$$

be respectively the supply and demand for consumption in period 2. Competitive prices p_1^* and $p_2^* = p_1^* + s$ solve $\tilde{S}(p_1^*) = \tilde{D}(p_1^*)$. From standard arguments, a competitive equilibrium exists, and it is unique under assumption (A). In our experiment, the supply shock treatment implements this case: a large increase in production costs between periods creates a gap $p_1^0 + s < p_2^0$, making storage profitable in the competitive equilibrium.

2.3 Expected price controls

We now consider a situation in which the price in period 2 is capped at some $\overline{p}_2 > 0$. If demand exceeds supply at $\overline{p}_2$, then the available supply is distributed randomly between those who demand the good, with equal probability. Given prices $p_1 \geq 0$ and $0 \leq p_2 \leq \overline{p}_2$ in periods 1 and 2, respectively, and a rationing probability ρ, the problem of the buyer i is

$$\max_{x_{i1} \in \{0,1,2\}, x_{i2} \in \{0,1\}} r_i(x_{i1} + \rho x_{i2}) - p_1 x_{i1} - p_2 \rho x_{i2} - s \max\{x_{i1} - 1, 0\},$$

and the problem of seller j is, as before,

$$\max_{y_{j1},y_{j2}\in\{0,1\}} (p_1 - c_{j1})y_{j1} + (p_2 - c_{j2})y_{j2}.$$

The market clearing and rationing conditions are

$$\begin{aligned}
&\int_{i\in B} x_{i1}di = \int_{j\in S} y_{j1}dj, \\
&\int_{i\in B} x_{i2}di \geq \int_{j\in S} y_{j2}dj \text{ with equality if } p_2 < \overline{p}_2, \\
&\rho = \frac{\int_{j\in S} y_{j2}dj}{\int_{i\in B} x_{i2}di} \text{ if } p_2 = \overline{p}_2 \text{ and } \int_{i\in B} x_{i2}di > 0, \text{ and } \rho = 1 \text{ otherwise.}
\end{aligned}$$

A rationing equilibrium is a demand vector (x_{i1}, x_{i2}) for each buyer $i \in B$, a supply vector (y_{j1}, y_{j2}) for each seller $j \in S$, a price vector (p_1, p_2) with $p_2 \leq \overline{p}_2$, and a rationing probability ρ such that buyers' demand and sellers' supply solve their respective problems, and the market clearing and rationing conditions hold. We say that the price ceiling is binding if $\bar{p}_2 < p_2^*$, and we say that rationing is effective if $\rho < 1$.

In the spirit of competitive equilibrium, we treat the probability ρ with which a buyer expects to be able to buy the good in period 2 if demanding the good as constant from the viewpoint of all buyers and equal for all buyers. We have:

Theorem 1 *In the expected price control case, there is a unique rationing equilibrium. Moreover, if the price ceiling is binding, in equilibrium:*

i) The supply in period 2 is strictly below the competitive supply and there is effective rationing.

ii) The price in period 1 plus the storage cost is strictly higher than the price ceiling in period 2. In contrast, in a competitive equilibrium, the price in period 1 plus the cost of storage is equal to the price in period 2 whenever there is storage.

iii) Only buyers with the highest reservations store in period 1. In contrast, in a competitive

equilibrium, buyers are either indifferent between storing in period 1 or buying in period 2, or prefer not to store.

Proof. If the price ceiling is binding, the supply in period 2 must be strictly below the maximum competitive supply. Additionally, $\rho < 1$ must hold; otherwise, the rationing equilibrium would also be a competitive equilibrium. This establishes part (i).

With respect to part (ii), a buyer prefers to store in period 1 rather than attempting to buy in period 2 if

$$r_i - (p_1 + s) \geq \rho \max(r_i - \overline{p}_2, 0),$$

with strict inequality for strict preference. If $\overline{p}_2 \geq p_1 + s$ and $\rho < 1$, it is better for all buyers to store in period 1 rather than attempting to buy in period 2, which contradicts $\rho < 1$, so we must have $\overline{p}_2 < p_1 + s$.

With respect to part (iii), using $\overline{p}_2 < p_1 + s$ we can write the condition for preferring to store as

$$(1 - \rho)r_i + \rho\bar{p}_2 \geq p_1 + s,$$

with strict inequality for strict preference. Thus, buyers with higher reservations store in a rationing equilibrium.

To prove existence and uniqueness, note that if the price ceiling is not binding, then the competitive equilibrium is the unique rationing equilibrium. Assume henceforth the price ceiling is binding; this implies $S_2(\overline{p}_2) < D(\overline{p}_2)$. Let $\rho_k = S_2(\overline{p}_2)/(D(\overline{p}_2) - \min(k, D(\overline{p}_2) - S_2(\overline{p}_2)))$ for $k \in [0, n_B]$. Intuitively, ρ_k represents the probability of successfully trading in period 2 for those buyers who decide to store instead of buying in period 2, given that a fraction k/n_B of the buyers choose to store. By definition, ρ_k increases in k.

We define the supply of stored goods as $\tilde{S}(p_1) = S_1(p_1) - D(p_1)$ as before, and the demand function for stored goods under rationing for $p_1 > \bar{p}_2 - s$ as

$$\hat{D}(p_1) = \begin{cases} 0 \text{ if } p_1 \geq (1 - \rho_0)r_0 + \rho_0\bar{p}_2 - s \\ \max\{i \in [0, n_B] : (1 - \rho_i)r_i + \rho_i\bar{p}_2 \geq p_1 + s\} \text{ otherwise.} \end{cases} \tag{1}$$

Intuitively, for every price $p_1 > \bar{p}_2 - s$ there is a marginal buyer who is indifferent to store or not, or every buyer prefers not to store. It is simple to verify that $\hat{D}(p_1)$ is continuous and strictly decreasing in p_1 for $\bar{p}_2 - s < p_1 < (1-\rho_0)r_0 + \rho_0\bar{p}_2 - s$, and $\lim_{p_1 \downarrow \bar{p}_2 - s} \hat{D}(p_1) = D(\bar{p}_2) - S_2(\bar{p}_2) = \tilde{D}(\bar{p}_2 - s)$.

Since the price ceiling is binding, we must have $\tilde{S}(\bar{p}_2 - s) < \tilde{D}(\bar{p}_2 - s)$. Hence either $(1-\rho_0)r_0 + \rho_0\bar{p}_2 - s \leq p_1^0$, in which case there is a unique rationing equilibrium with zero storage for $p_1 = p_1^0$ and $\rho = \rho_0$, or $(1-\rho_0)r_0 + \rho_0\bar{p}_2 - s > p_1^0$, in which case there is a unique rationing equilibrium for $p_1^0 < p_1 < \bar{p}_2 - s$ such that $\hat{D}(p_1) = \tilde{S}(p_1)$ and $\rho = \rho_x < 1$ for $x = \hat{D}(p_1)$. ■

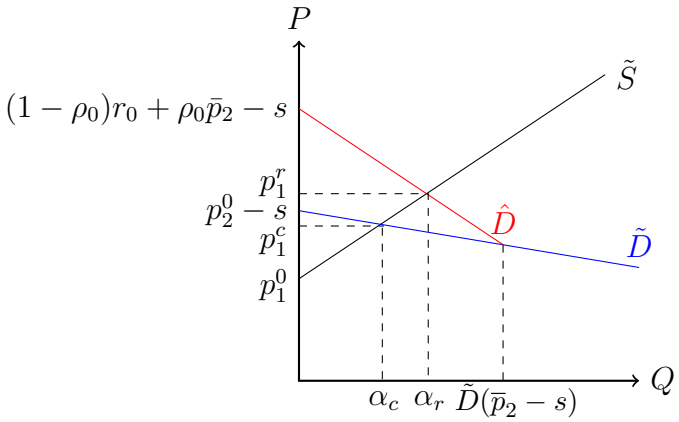


Figure 1: Market for storage in competitive and rationing equilibrium

Figure 1 illustrates the market for storage under competitive conditions and under rationing when a price cap $\bar{p}_2$ is expected. For $p_1 > \bar{p}_2 - s$, demand under rationing $\hat{D}$ is above competitive demand $\tilde{D}$. This indicates a higher willingness to pay for additional widgets in the first period given the expected shortage. As more agents buy in advance, the probability of being left out of the market in the second period declines. At the price $p_1 = \bar{p}_2 - s$ where the mismatch of supplied and demanded quantities at the price cap is completely satisfied, $\hat{D}$ is equal to $\tilde{D}$. Note that additional storage occurs compared to the competitive equilibrium ($\alpha_r > \alpha_c$), yet a certain level of shortage still remains ($\alpha_r < \tilde{D}(\bar{p}_2 - s) = D(\bar{p}_2) - S_2(\bar{p}_2)$).

2.4 Expected liberalization

We consider now a scenario with an initial price cap, $p_1 = \bar{p}_1$, and an anticipated market liberalization where p_2 is freely determined by the market. Here, buyers are incentivized to buy early at regulated prices when expecting price increases during the market liberalization.

Consumers can demand zero, one, or two units of the good in period 1, and that there is a probability ρ of successfully trading in the market due to excess demand at the regulated price.[4] If the consumer successfully purchases two units in period 1, then the consumer stores the additional unit and does not buy the good in period 2.

The consumer problem is now given by:

$$\max_{x_{i1}\in\{0,1,2\},x_{i2}\in\{0,1\}} \rho x_{i1}(r_i - \overline{p}_1) - \rho \max(x_{i1}-1,0)s + (1-\rho\max(x_{i1}-1,0))\, x_{i2}(r_i - p_2),$$

where $x_{i1} \in \{0,1,2\}$, $x_{i2} \in \{0,1\}$, and $\rho \in (0,1]$, and the seller's problem j remains the same as before. In the specification above, the consumer buys a unit in the second period only if attempting to buy fewer than two units in the first period, or if attempting to buy two units in the first period but being unsuccessful due to rationing.

We assume that each buyer (correctly) anticipates that the probability of being able to purchase in the first period is constant and the same for all buyers. The market clearing and rationing conditions are

$$\begin{aligned}
&\int_{i\in B} x_{i1}di \geq \int_{j\in S} y_{j1}dj \;\text{ with equality if }\; p_1 \leq \overline{p}_1,\\
&\int_{i\in B} x_{i2}di = \int_{j\in S} y_{j2}dj,\\
&\rho = \frac{\int_{j\in S} y_{j1}dj}{\int_{i\in B} x_{i1}di} \text{ if } p_1 = \overline{p}_1 \text{ and } \int_{i\in B} x_{i1}di > 0, \text{ and } \rho = 1 \text{ otherwise.}
\end{aligned}$$

A rationing equilibrium is now a demand vector (x_{i1}, x_{i2}) for each buyer $i \in B$, a supply vector (y_{j1}, y_{j2}) for each seller $j \in S$, a price vector (p_1, p_2), where $p_1 \leq \overline{p}_1$, and a rationing

[4] This anonymous rationing rule adheres to the conditions proposed by Green (1980), and is exactly in alignment with the rule we used in our laboratory implementation.

probability ρ, such that buyers' demand and sellers' supply solve their respective problems, and market clearing and rationing conditions hold.

Theorem 2 *In the expected liberalization case, there is a unique rationing equilibrium. Moreover, if the price ceiling is binding, in equilibrium:*

i) The supply in the period 1 is strictly below the competitive supply and there is effective rationing.

ii) If there is storage, the price in period 2 is strictly lower than the myopic price.

iii) Either only buyers with the highest reservations attempt to buy for storage in period 1, or buyers who store are indifferent between storing or not.

Proof. Part (i) is immediate. For part (ii), note that if there is storage then demand in period 2 is below the myopic demand while supply is equal to the myopic supply, leading to a lower than myopic price.

For part (iii), from the objective function of buyers we get that the utility of $x_{i1} = 0, 1, 2$ are given respectively by $\max(0, r_i - p_2)$, $\rho(r_i - \bar{p}_1) + \max(0, r_i - p_2)$, and $2\rho(r_i - \bar{p}_1) - \rho s + (1-\rho)\max(0, r_i - p_2)$. Hence, attempting to buy for storage ($x_{i1} = 2$) is optimal if $r_i - \max(0, r_i - p_2) \geq \bar{p}_1 + s$ with strict preference for strict inequality. Thus, agents only store if $p_2 \geq \bar{p}_1 + s$; moreover, if $p_2 > \bar{p}_1 + s$ then every buyer such that $r_i > \bar{p}_1 + s$ strictly prefers to store, while if $p_2 = \bar{p}_1 + s$ every buyer such that $r_i \geq \bar{p}_1 + s$ is indifferent between storing or not.

To prove existence and uniqueness, note that if the price ceiling is not binding, then the competitive equilibrium is the unique rationing equilibrium. Assume henceforth the price ceiling is binding; this implies $S_1(\bar{p}_1) < D(\bar{p}_1)$. If $p_2^0 \leq \bar{p}_1 + s$, then there is a unique equilibrium with rationing with $p_2 = p_2^0$ and $\rho = S_1(\bar{p}_1)/D(\bar{p}_1)$; in this case there is no storage in equilibrium.

Assume henceforth $p_2^0 > \bar{p}_1 + s$. If

$$\left(1 - \frac{S_1(\bar{p}_1)}{D(\bar{p}_1) + D(\bar{p}_1 + s)}\right) D(\bar{p}_1 + s) > S(\bar{p}_1 + s),$$

then the unique rationing equilibrium is given by $p_2^r \in (\bar{p}_1 + s, p_2^0)$ such that

$$\left(1 - \frac{S_1(\bar{p}_1)}{D(\bar{p}_1) + D(\bar{p}_1 + s)}\right) D(p_2^r) = S(p_2^r)$$

and $\rho = S_1(\bar{p}_1)/(D(\bar{p}_1) + D(\bar{p}_1 + s))$. In this case there is positive storage, buyers who store strictly prefer so, and some of the buyers who store prefer not to buy in the second period if unsuccessful.

If instead

$$\left(1 - \frac{S_1(\bar{p}_1)}{D(\bar{p}_1) + D(\bar{p}_1 + s)}\right) D(\bar{p}_1 + s) \leq S(\bar{p}_1 + s),$$

then there is a unique rationing equilibrium given by $p_2^r = \bar{p}_1 + s$ and a fraction $0 < \alpha \leq 1$ of buyers with valuations $r_i \geq \bar{p}_1 + s$ attempting to buy in the first period so that

$$\left(1 - \frac{S_1(\bar{p}_1)\alpha}{D(\bar{p}_1) + \alpha D(\bar{p}_1 + s)}\right) D(\bar{p}_1 + s) = S(\bar{p}_1 + s)$$

and $\rho = S_1(\bar{p}_1)/(D(\bar{p}_1) + \alpha D(\bar{p}_1 + s))$. In this case there is positive storage, buyers who store are indifferent, and some of the buyers who store prefer not to buy in the second period if unsuccessful. ■

In both cases, expected price controls and liberalization, storage allows consumers to shift purchases from period 1 to period 2. The key difference lies in the motive: under expected price controls, buyers store to avoid future scarcity; under expected liberalization, they store to avoid future price increases.

3 Experimental design

3.1 Hypotheses

This section outlines the hypotheses of our previous theoretical framework. We will test these through our lab experiment, examining market outcomes and the effects of price distortions and forward-looking behavior.

Hypothesis 1 *In a competitive market with storage, buyers are indifferent between storing a good and buying it in the second period, that is, the price of the first period, combined with storage costs, is equal to the price of the second period. There is no rationing in equilibrium.*

Hypothesis 2 *In a market that experiences price distortions but expects liberalization in the future (price cap in $t = 1$), the rationing equilibrium maintains the following conditions:*

a) The demand in the first period is strictly higher than in the competitive equilibrium, resulting in shortages.

b) All buyers with valuations greater than $\frac{\bar{p}_1 - \rho(p_2 - s)}{1-\rho}$ and $\bar{p}_1 + \rho s$ attempt to buy for storage.

c) If there is storage, the price in the second period is strictly lower than the myopic price and may or may not be greater than or equal to the regulated price plus the storage cost.

Hypothesis 3 *In a market that expects price distortions in the future (price cap in $t = 2$), the rationing equilibrium sustains the following conditions:*

a) High-reservation buyers store in period 1, while low-reservation buyers purchase the good in period 2 at the price ceiling $\overline{p}_2$.

b) The price in period one, when added to the cost of storage, is strictly larger than the price ceiling in period 2.

c) The price in period 1 is larger than or equal to the competitive price in period 1.

d) The quantity of goods stored in the rationing equilibrium is greater than or equal to the quantity stored in the competitive equilibrium.

3.2 Experiment

The experiment consists of trading a durable good over two periods. In the initial period, buyers can opt to store goods at a certain cost for use in the subsequent period. Those who

acquired an extra unit of the good in the first period are not permitted to partake in the second period's market.

We utilize a between-subjects approach, randomly assigning different treatments to participants across sessions. Each participant receives a budget of 200 lab currency units. We assigned valuations randomly to ensure equal expected earnings for everyone throughout the experiment.

In terms of structure, each session forms groups of eight participants, and the entire experiment covers 10 rounds. Each round includes two trading periods, consistent with our experimental model. For our analysis, we treat pairs of group rounds as a single market. In each market, there are eight human buyers and nine "robot" sellers. These robot sellers are programmed to adhere to equilibrium strategies. The use of automated sellers is a common approach in experimental markets (Cason and Friedman, 1997) and allows us to isolate buyer behavior, eliminating seller-side behavioral effects and reducing the noise from strategic interaction on both sides of the market. A detailed set of instructions and experiment materials are available in the online appendix.

3.3 Market institution

We use a single call market (SCM) institution. SCM asks each buyer i to submit their maximum acceptable price, bid b_i, and to each seller to submit their minimum acceptable sale price, ask a_j. The market is cleared at a single equilibrium price p^*. This particular market institution reduces behavioral noise, guarantees a unique equilibrium price, and human subjects seem to behave more efficiently when interacting with automated agents (Cason and Friedman, 1997).

3.4 Implementation of the model

We utilize three different treatments for this experiment. In all treatments, buyers' valuations remain constant across both periods, while sellers' production costs increase fourfold in the second period ($c_2(i) = 4 \times c_1(i)$). The treatments differ in whether and when a price ceiling

is imposed.

1. **Supply shock**: This treatment serves as the baseline. There is no price cap in either period. Sellers' costs are four times greater in the second period.

   Demand: $r(i) = [50, 50, 70, 90, 110, 140, 140, 200]$

   Supply: $c_1(i) = [5, 20, 20, 30, 50, 55, 60, 80, 100]$ and
   $c_2(i) = [20, 80, 80, 120, 200, 220, 240, 320, 400]$.

   In the myopic equilibrium, the quantity traded in the first period is $q_1 = 6$, with the corresponding price falling within the range of $[55, 60]$. Similarly, the equilibrium quantity traded in the second period is $q_2 = 3$, and the corresponding price is in the range of $p_2 \in [110, 120]$.

   Notice that $p_2 > p_1 + s$ indicates an incentive to store goods in period 1. As a result, the predicted competitive equilibrium leads to the following outcomes:

   - Equilibrium probability: $\rho = 1$ (no rationing in equilibrium).
   - Demand for storage: $\tilde{D} = [50, 70, 70, 70, 80, 110]$
   - Supply for storage: $\tilde{S} = [60, 70, 80, 90, 100, 110, 140, 140, 200]$
   - In the competitive equilibrium, the quantity traded in the first period can be either $q_1^* = 7$ or $q_1^* = 8$, with the corresponding price falling within the $[70, 80]$ range. $\alpha^* = 2$ units are put in storage. For the second period, the equilibrium quantity traded is $q_2^* = 3$, and the corresponding price is in the range of $[80, 90]$.

[Figure 2 about here]

2. **SS + Price Cap**: In this treatment, we introduce a price cap regulation in the second period. Buyers' valuations remain constant across both periods, and sellers' costs are four times greater in the second period, as in the supply shock treatment. In addition, a price ceiling is imposed in the second period.

Demand: $r(i) = [50, 50, 70, 90, 110, 140, 140, 200]$

Supply: $c_1(i) = [5, 20, 20, 30, 50, 55, 60, 80, 100]$ and
$c_2(i) = [20, 80, 80, 120, 200, 220, 240, 320, 400]$.

The price ceiling, $\bar{p}_2$, is set at 50 in period 2. After the supply shock, only one seller has a cost below the ceiling ($c_{j2} = 20 \leq 50$), so $S_2(\bar{p}_2) = 1$ while $D(\bar{p}_2) = 8$. This severe shortage triggers a strong demand for storage with a cost of $s = 10$. As a result, the demand for storage is given by $\tilde{D} = [171.62, 170.62, 116.43, 114.17, 87.0, 68.75, 51.67, 37.5]$, while the supply for storage is $\tilde{S} = [60, 70, 80, 90, 100, 110, 140, 140, 200]$. Therefore, the predicted market equilibrium is as follows:

- Equilibrium probability of acquiring a good in period 2 is $\rho \in [0.17, 0.20)$.
- In the rationing equilibrium, the quantity traded in the first period is $q_1^* = 8$, among them $\alpha^* = 3$ are put in storage, with the corresponding price p_1^* falling within the range $[88, 90]$. In the second period, the traded quantity is $q_2^* = 1$ (only the lowest-cost seller can sell below the ceiling), with the corresponding price $p_2^* = \bar{p}_2 = 50$.

[Figure 3 about here]

3. **Liberalization**: In this treatment, we introduce a price cap regulation in the first period that is removed in the second period. Buyers' valuations remain constant across both periods, and sellers' costs are four times greater in the second period, as in the other treatments.

   Demand: $r(i) = [50, 50, 70, 90, 110, 140, 140, 200]$

   Supply: $c_1(i) = [5, 20, 20, 30, 50, 55, 60, 80, 100]$ and
   $c_2(i) = [20, 80, 80, 120, 200, 220, 240, 320, 400]$.

   The price ceiling, $\bar{p}_1$, is set at 50 in period 1, and is binding, resulting in a shortage during that period. The cost of storage is set at $s = 10$. The predicted market

equilibrium is as follows.

- The supply at the ceiling in period 1 is $S_1(\bar{p}_1) = 5$ (five sellers have costs ≤ 50), while the demand is $D(\bar{p}_1) = 8$. The equilibrium probability of acquiring a good in period 1 is $\rho = 5/14$, as 6 buyers with valuations ≥ 70 demand 2 widgets and 2 buyers with valuations of 50 demand 1 widget each.
- In the rationing equilibrium, the quantity traded in the first period is $q_1^* = 5$, with the corresponding price $p_1 = 50$, the price ceiling.
- In the second period, the traded quantities are $q_2^* \in [3, 5]$, with the corresponding price $p_2^* \in [70, 120]$, reflecting the supply shock and the effect of any storage from the first period.

[Figure 4 about here]

We administer this experiment using an online platform developed in oTree (Chen et al., 2016). At George Mason University, we organized 19 sessions, each involving 8 or 16 students, all acting as buyers. Our final sample consists of 264 participants (88 per treatment, 11 groups each). Of these, 146 are male, 111 are female, 4 identify as non-binary, and 3 chose not to reveal their gender. About 59% of participants are enrolled in STEM programs. Table 1 details the results of our post-experiment demographic survey.

[Table 1 about here]

4 Experimental results

4.1 Overview

Our analysis relies on nonparametric tests (Wilcoxon–Mann–Whitney, Pearson's chi-squared) and one-sample t-tests at the market level. With 11 independent groups per treatment and

55 market observations per treatment in the last 5 rounds, regression-based approaches would face low statistical power and difficulty satisfying distributional assumptions. Our between-subjects design and the use of group-level averages as independent observations ensure that the tests we report are appropriate for the sample size and data structure.[5]

Free market (Supply Shock)

H1: *The price in the first period, combined with storage costs, equals the price in the second period, and there is no rationing.*

Results: In the supply shock treatment, some buyers had an incentive to buy early because production costs increase fourfold in the second period. Trading prices during the experiment align with the competitive equilibrium: the first-period price averages 73.67, within the predicted range of [70, 80], and the second-period price averages 88.15, within the predicted range of [80, 90]. We confirm that there is no rationing, consistent with the theory. However, the no-arbitrage condition $p_1 + s = p_2$ holds only approximately: the average excess gap $p_2 - p_1 - s = 4.47$ is statistically different from zero ($t = 2.32$, $p = 0.024$, $N = 55$). This residual gap suggests that storage arbitrage does not fully close the price spread: participants store more than predicted, yet their first-period bids remain too low to eliminate the arbitrage opportunity.

Liberalization

H2a: *The demand in the first period is strictly higher than in the competitive equilibrium, resulting in shortages.*

Results: We confirm this hypothesis. Shortages are near zero in all periods without a price cap. In the liberalization treatment, where a price cap binds in the first period, we find an average shortage rate of 25%. As shown in Figure 12, shortages between 50% and 60% are the most frequent outcome, consistent with our theoretical predictions. Notably, the shortage rate increases over rounds, from 16% in round 1 to 36% in round 10, as participants learn that the price cap binds and that buying at the regulated price yields large rents relative to the free second-period market.

[5]Throughout the analysis, unless otherwise noted, we use data from the last 5 rounds (rounds 6–10), when participants have had sufficient experience with the market institution and incentives.

H2b: *All buyers with valuations greater than* $\frac{\bar{p1}-\rho(p_2-s)}{1-\rho}$ *and* $\bar{p_1}+\rho s$ *attempt to buy for storage*

Results: We have evidence to confirm this hypothesis. When considering the last 5 rounds, players who attempt to buy two goods are likely to have valuations higher than 50, the threshold value given the parameters of the implementation. Players with the lowest valuation are less likely to attempt to store goods.

H2c: *If there is storage, the price in the second period is strictly lower than the myopic price and may or may not be greater than or equal to the regulated price plus the storage cost.*

Results: We confirm this hypothesis. The second-period price averages 109.04, which falls within the predicted range of [70, 120] and is below the myopic equilibrium of [110, 120]. Storage from the first period shifts effective demand downward in the second period, reducing the price relative to what would prevail without storage. In practice, the liberalized price represents a dramatic increase from the first-period cap of 50. Buyers who stored at the regulated price paid a total of 60 (price plus storage cost) for goods that would cost 109 in the free market, capturing a rent of approximately 49 per stored unit. This price spike after liberalization is the central feature of this treatment: the sequence of price controls followed by market liberalization produces the largest price increase and the most pronounced redistribution of surplus in our experiment.

SS + Price Cap

H3a: *high-reservation buyers store in period 1, while low-value buyers purchase the good in period 2 at the price ceiling* $\overline{p_2}$.

Results: We have sufficient evidence to confirm this hypothesis. There is a statistical difference in valuations between participants who chose to buy two goods in the first period and those who did not. Moreover, buyers with higher values were more inclined to buy two goods. In other treatments, however, buyers did not appear to base their storage decisions on their valuations.

H3b: *The price in period one, when added to the cost of storage, is strictly larger than*

the price ceiling in period 2.

Results: Our data support this hypothesis. With the introduction of price cap regulation at $t = 2$, the average price in the first period was significantly higher than the price ceiling set for the subsequent period. However, while the average price during the first period aligns with theoretical predictions, it falls below equilibrium in the second period.

H3c: *The price in period 1 is greater than or equal to the competitive price.*

Results: In the SS + Price Cap treatment, the first-period price averages 74.35, which is above the myopic equilibrium of [55, 60] but below the rationing equilibrium prediction of [88, 90]. The price in the first period exceeds what would prevail without storage incentives. However, Table 4 shows that the difference in first-period prices between the supply shock and the SS + Price Cap treatments is not statistically significant.

H3d: *The quantity of goods stored in the rationing equilibrium is larger than or equal to the quantity stored in the competitive equilibrium.*

Results: In the SS + Price Cap treatment, participants stored 2.38 units per market, which exceeds the competitive equilibrium of 2 units but falls short of the rationing equilibrium prediction of 3 units. This is consistent with the broader pattern observed throughout our experiment: participants store in the right direction but less aggressively than predicted, as if they underestimate the severity of future rationing.

4.2 Bidding strategy

We start by examining the bidding strategies adopted by the participants in each treatment.[6] Figures 5 and 6 illustrate the average price offered in each period based on the valuation assigned during the experiment. From these graphs, it is evident that participants generally offer prices below their valuation, as expected. The prices offered in the initial five rounds are slightly higher than those in the final five rounds, indicating adjustments. However, there

[6] A coding error in our oTree application prevented us from recording the period 1 bid prices for 4 of the 11 supply shock groups (32 participants across all 10 rounds, or 320 bid-price observations). The early version of the application overwrote the period 1 bid price with the period 2 bid price. This error does not affect transaction prices, quantities, or any other variable, which were recorded correctly for all groups and rounds. Bid-price figures and analyses for the supply shock treatment use only the 7 unaffected groups.

are no significant differences between the outcomes of the first and last 5 rounds.

Furthermore, although bid prices tend to gravitate near the equilibrium, there is a tendency to overbid in the first period and underbid in the second, especially in the supply shock and SS + Price Cap treatments.

In contrast, the bid quantities consistently exceeded optimal levels. Our findings reveal a substantial excess of demand at all valuation levels, resulting in an inefficient allocation of widgets. Consequently, participants' valuations did not have a significant influence on the final quantity of widgets they secured during the experiment.

Figure 7 presents the average bidding strategy for all treatments. In particular, buyers at all valuation levels attempted to buy on average the same amount of goods in the first period. Table 2 shows that there are no statistical differences in the valuation of those who attempted to buy one or two goods, except in the SS + Price Cap treatment. In that case, we confirm the hypothesis that buyers with higher valuations have a greater likelihood of buying for storage. In the liberalization treatment, a similar trend is observed among high-value buyers, which is contrary to our predictions. However, these variations are only statistically significant in one of our differences in distribution tests.

[Figure 5 about here]

[Figure 6 about here]

[Figure 7 about here]

[Table 2 about here]

4.3 Market prices

Table 3 displays the average trading price in all treatments and trading periods. Prices fall within the expected range in the supply shock and liberalization treatments. In the

SS + Price Cap treatment, the first-period price (74.35) is above the myopic equilibrium but below the rationing equilibrium prediction of [88, 90].

In the supply shock treatment, prices in both periods align with the competitive equilibrium. The first-period price averages 73.67 (predicted: [70, 80]) and the second-period price averages 88.15 (predicted: [80, 90]).

In the SS + Price Cap treatment, the second-period price matches the binding price cap at 49.91. However, the first-period price is lower than predicted by the rationing equilibrium, suggesting that participants bid too little for storage.

In the liberalization treatment, the first-period price matches the binding cap at 49.96. Once the market is liberalized in the second period, the price jumps to 109.04 — more than double the first-period cap. This falls within the predicted range of [70, 120], but the magnitude of the increase illustrates the consequences of the price-control-then-liberalize sequence. Second-period quantities drop to 2.75 (vs a myopic prediction of 3), as the free market clears at a much higher price with fewer participants.

Figure 8 illustrates the distribution of trading prices for each treatment and trading period over the last 5 rounds.[7] In most markets, prices consistently align with the equilibrium range. The most frequent price tends to be close to the optimal pricing range.

Table 4 compares the first-period price and the amount of stored goods across all three treatments. The difference in first-period prices between the supply shock and SS + Price Cap treatments is not statistically significant. Storage is also similar between the two treatments (2.36 vs 2.38 units per market, $p = 0.91$), despite different theoretical predictions (2 vs 3). Both treatments show substantially more storage than the liberalization treatment (1.15 units per market), where storage is motivated by rent-seeking at the regulated price rather than precautionary hoarding.

[Table 3 about here]

[Figure 8 about here]

[7]Tables 10, 11, 12, and 13 in the appendix replicate the main results tables using all 10 rounds. The results are qualitatively identical.

It is worth noting that, except for the liberalization treatment, we do not observe a prolonged learning trend or convergence path to equilibrium. Figures 17 and 15 depict the progression of average trading prices across the experiment rounds, revealing that learning took place between the first two rounds and remained nearly stagnant afterward. Furthermore, Table 6 confirms that the quantity of goods purchased remains stable between the initial and concluding 5 rounds. Only the liberalization treatment shows a trend towards the equilibrium, and the observed differences are statistically significant.

[Table 4 about here]

4.4 Consumption and hoarding

Table 5 displays the average quantities traded in each market, separated by treatments and trading periods. In the supply shock treatment, the first-period quantity (7.04) is close to the predicted range of [7, 8], while the second-period quantity (2.05) falls below the predicted 3 units, likely due to excessive storage. In the liberalization treatment, traded quantities are close to the predicted values. In the SS + Price Cap treatment, the first-period quantity (7.11) is near the predicted 8 units, and the second-period quantity matches the prediction of 1 unit exactly.

As indicated previously, we do not observe a clear convergence path to the equilibrium. The results in Table 6 show that the amount of goods purchased is similar between the first and last five rounds for all treatments, except for the liberalization treatment.

[Table 5 about here]

[Table 6 about here]

Traded quantities often do not match the competitive equilibrium. Figure 9 displays the distribution of quantities traded during the last 5 rounds of the experiment. As highlighted in our earlier analysis of bidding strategies, it is worth noting that while buyers often prioritize

price levels in their bids, they frequently neglect the broader market equilibrium implications when choosing quantities. Moreover, Figure 10 suggests that buyer's valuation is not consistently the primary determinant of demand choices. Individuals with high valuations are often sidelined in the market. Surprisingly, low value buyers, who do not appear to have any advantage from buying goods, succeed in obtaining the goods across all treatments.

[Figure 9 about here]

[Figure 10 about here]

Figure 11 shows the number of goods stored in the first period. In the supply shock treatment, participants store slightly more than predicted (2.36 vs 2 units per market). In the SS + Price Cap treatment, participants store less than predicted (2.38 vs 3 units per market). The two treatments produce nearly identical storage levels despite different theoretical predictions, suggesting that participants respond similarly to storage incentives regardless of whether the motivation is competitive arbitrage or defensive hoarding. In the liberalization treatment, participants store on average 1.15 units per market. Here, storage is motivated not by avoiding future scarcity but by capturing the rent between the regulated first-period price and the free second-period price, a form of rent-seeking rather than defensive storage. Storage remains stable throughout the rounds of the experiment across all treatments.

[Figure 11 about here]

4.5 Shortage and Efficiency

Prices serve as signals that help people understand what is in demand and what is available. Efficiency is maximized when supply and demand determine prices. When external forces impose distortions on the price mechanism, these signals become distorted, misleading agents and causing inefficiencies. In this section, we analyze the impact on shortage and efficiency across our three treatments.

We argue that storage serves as a defense mechanism that improves efficiency when there are price distortions. If distortions are anticipated, as in the SS + Price Cap treatment, consumers can store goods, reducing the deadweight loss. On the other hand, when prices are in disequilibrium but liberalization is expected, individuals have an incentive to capture rents. In the liberalization treatment, some buyers try to buy in excess to take advantage of the lower prices in the first period. In the supply shock treatment, we do not anticipate any rationing.

Table 7 shows the average shortage rate for each treatment and period, using all 10 rounds.[8] We calculate the shortage rate as the share of unsatisfied demand among buyers who bid at or above the transaction price. The results confirm our main theoretical predictions. In the supply shock treatment, shortage rates are near zero in both periods (4% in period 1, 3% in period 2), consistent with a free market without binding price controls.

In the liberalization treatment, where the price cap binds in the first period, we observe a 25% average shortage rate in period 1. However, rates between 50% and 60% are the most frequently recorded shortage rate, as shown in Figure 12. The shortage worsens over rounds as participants learn to exploit the regulated price: the rate rises from 16% in round 1 to 36% in round 10. This dynamic has an important policy implication: the longer a price cap remains in place, the more agents learn to arbitrage it, and the worse the shortage becomes. Once the market is liberalized in the second period, shortage drops to 2%, but at the cost of a sharp price increase.

In the SS + Price Cap treatment, the first period operates as a free market and shows an 8% shortage rate. The second period, where the price cap binds, shows a 60% shortage rate. Storage helps reduce the shortage below the static prediction, but does not eliminate it.

[Table 7 about here]

[8] We use all 10 rounds for the shortage analysis rather than the last 5 for two reasons. First, the worsening-over-rounds dynamic in the liberalization treatment is itself a central finding that requires the full sample to document. Second, the coding error that affected period 1 bid prices in 4 supply shock groups reduces the available sample for that treatment; using all rounds preserves statistical power. As documented in Table 6, quantities are stable across early and late rounds, so this choice does not bias the shortage estimates.

[Figure 12 about here] To analyze efficiency, we measure total surplus: the sum of consumption values minus production costs and storage costs. Since sellers are automated, this captures the full welfare generated by the market. We define the efficiency ratio κ for market m as:

$$\kappa_m = \frac{W_m}{W_m^*}, \qquad W_m = \sum_{i \in B} r_i q_{im} - \sum_{j \in S} c_j y_{jm} - \alpha_m s$$

where $r_i q_{im}$ is the consumption value of buyer i, $c_j y_{jm}$ is the production cost of seller j, α_m is the number of stored units, and $s = 10$ is the per-unit storage cost. The transaction price cancels out of total surplus because it is a transfer between buyers and sellers. The benchmark W_m^* is the theoretical equilibrium total surplus: the competitive equilibrium for the supply shock treatment and the rationing equilibrium for the price cap treatments.

Table 8 reports both total surplus and the efficiency ratio. The supply shock treatment generates the highest total surplus (701 LCU per market), followed by the SS + Price Cap treatment (652 LCU) and the liberalization treatment (609 LCU). Notably, the efficiency ratios are similar across treatments: 77% for the supply shock, 75% for SS + Price Cap, and 77% for liberalization. Each treatment loses roughly a quarter of its theoretical surplus, but because the theoretical benchmarks differ substantially ($W^* = 915$, 864, and 793, respectively), the absolute welfare losses under price controls are large.

All three treatments achieve similar efficiency ratios relative to their treatment-specific theoretical predictions, as expected: each market institution generates roughly three-quarters of its theoretical surplus. However, the realized total surplus is highest in the free market and lowest under anticipated liberalization, where rent-seeking storage predominates. In the supply shock treatment, participants store slightly more than predicted (2.36 vs 2 units per market), while the second-period quantity falls below equilibrium (2.05 vs 3). In the SS + Price Cap treatment, participants under-store relative to the rationing equilibrium (2.38 vs 3) and bid too little for storage. Despite these opposing deviations from theory, storage levels converge to nearly identical values across the two treatments, suggesting a

common behavioral anchor around 2.4 units regardless of the underlying incentive structure.

The liberalization treatment produces the lowest total surplus of all three treatments. This outcome results from the combination of three forces. First, the binding price cap in period 1 causes rationing, randomly excluding some buyers (including high-value ones) from the market. Second, buyers who manage to store at the regulated price capture large rents (paying 60 for goods worth 109 in the free market), while buyers rationed out of period 1 face the full liberalized price. Third, the second-period quantity collapses to 2.75 units (vs a myopic prediction of 3), as the market clears at a much higher price.

It is worth noting that we are not considering the effect of risk aversion or preferences for early resolution but the pure effect of the price mechanism. Our benchmarks do not necessarily reflect the true maximization of utility.

[Table 8 about here]

Figure 13 compares the efficiency ratio across rounds for each treatment. The supply shock treatment shows the clearest learning trajectory, rising from 59% in round 1 to 85% by round 10. The SS + Price Cap and liberalization treatments fluctuate around their respective means with no clear trend, suggesting that the efficiency losses from price controls are persistent rather than diminishing with experience.

[Figure 13 about here]

5 Conclusions

In this paper, we model hoarding as a rational response to price distortions and test the predictions experimentally. Our theory accurately predicts prices and shortage patterns across three treatments: shortages arise only in periods with binding price caps, and periods without controls show near-zero shortage rates. We distinguish between defensive storage, where buyers hoard to insure against future scarcity, and rent-seeking storage, where buyers

hoard to capture the gap between regulated and market prices. The data confirm this distinction.

Our experiments reveal a consistent behavioral pattern: participants store fewer units than predicted and bid too little for them. In the SS + Price Cap treatment, participants stored 2.4 units per market instead of the predicted 3, and the first-period price averaged 74 instead of the predicted [88, 90]. They hoard, but as if they underestimate the severity of future rationing and expect to access the constrained market anyway. In terms of total surplus, the free market generates the highest welfare (701 LCU per market), while both price-control treatments produce lower surplus: 652 LCU for the SS + Price Cap treatment and 609 LCU for liberalization. All three treatments achieve similar efficiency ratios relative to their treatment-specific benchmarks (75–77%), but the absolute welfare difference is substantial.

The liberalization treatment illustrates the costs of lifting price controls. Once the cap is removed, the price more than doubles — from 50 to 109 — and the first-period shortage worsens over rounds (from 16% in round 1 to 36% in round 10) as participants learn to exploit the regulated price. This suggests that price controls become harder to sustain the longer they are in place, as agents learn to arbitrage the gap between the regulated and market price. The combination of first-period rationing, rent-seeking storage, and a sharp post-liberalization price spike makes this the worst-performing scenario in our experiment.

Several factors may contribute to the gap between predicted and observed behavior, including underestimation of equilibrium consequences (Dal Bó et al., 2018), preferences for early resolution of uncertainty (Kreps and Porteus, 1978), and cognitive biases or risk aversion (Ellsberg, 1961; Kahneman et al., 1991; Alan and Ertac, 2015; Sunstein, 2000; de Clippel and Rozen, 2023). More research is needed to understand why participants store in the right direction but fail to bid accordingly, and to explore whether these dynamics extend to richer settings with repeated liberalization episodes.

Table 1: Participants

	Supply Shock	**SS + Price Cap**	**Liberalization**
No. of participants	88	88	88
Gender			
Man	58%	52%	56%
Woman	40%	44%	42%
Non-binary	0%	0%	0%
NR	1%	2%	0%
Major			
STEM	58%	60%	59%
Non-STEM	26%	28%	19%
Economics	3%	1%	3%
NR	12%	10%	18%
Level of studies			
Graduate	53%	68%	76%
Undergraduate	44%	28%	20%
NR	2%	3%	3%

Table 2: Difference in player's valuation for those who bid for 1 or 2 widgets

Player's valuation	Bidding Quantity = 1	Bidding Quantity = 2	Total	Pearson's Chi-2 Test	Wilcoxon Mann Whitney Test
Supply Shock					
50	46	54	100		
70	29	24	53		
90	19	34	53	6.3251	-1.2290
110	26	27	53	Pr = 0.276	Pr = 0.210
140	41	67	108		
200	22	31	53		
SS + Price Cap					
50	43	58	101		
70	22	29	51		
90	23	29	52	6.7010	-1.0013
110	15	40	55	Pr = 0.244	Pr = 0.307
140	35	71	106		
200	23	30	53		
Liberalization					
50	48	57	105		
70	29	26	55		
90	24	29	53	7.8241	-1.8457
110	16	37	53	Pr = 0.166	Pr = 0.060
140	42	68	110		
200	20	35	55		

Note: The table uses the data for the last 5 rounds in each treatment

Table 3: Market Price

Treatment	Period	N	Price	se	95% CI	Equilibrium
Supply Shock	1	55	73.67	1.44	[70.85, 76.50]	[70, 80]
	2	55	88.15	1.59	[85.03, 91.26]	[80, 90]
SS + Price Cap	1	55	74.35	1.49	[71.42, 77.27]	[88, 90]
	2	55	49.91	0.09	[49.73, 50.09]	[50, 50]
Liberalization	1	55	49.96	0.04	[49.89, 50.03]	[50, 50]
	2	55	109.04	3.51	[102.16, 115.91]	[70, 120]

Note: Uses data from last 5 rounds

Table 4: Differences in market price and stored goods

	Supply Shock	SS + Price Cap	Liberalization	T-Test	Mann-Whitney
Num. Markets	55	55	55	-	-
Price in $t = 1$	73.67	74.35	49.96	-0.3244	1493.5000
	(1.441)	(1.491)	(0.036)	Pr = 0.7462	Pr = 0.9112
Stored Goods	2.36	2.38	1.15	-0.1187	1491.0000
	(0.108)	(0.109)	(0.102)	Pr = 0.9058	Pr = 0.8920

Note: Uses data from last 5 rounds. T-test and Mann-Whitney compare Supply Shock vs SS + Price Cap.

Table 5: Market Quantities

Treatment	Period	N	Q	se	95% CI	Equilibrium
Supply Shock	1	55	7.04	0.10	[6.84, 7.23]	[7, 8]
	2	55	2.05	0.11	[1.83, 2.28]	[3, 3]
SS + Price Cap	1	55	7.11	0.10	[6.92, 7.30]	[8, 8]
	2	55	1.00	0.00	[1.00, 1.00]	[1, 1]
Liberalization	1	55	4.93	0.04	[4.86, 5.00]	[5, 5]
	2	55	2.75	0.12	[2.52, 2.97]	[3, 5]

Note: Uses data from last 5 rounds

Table 6: Sold widgets distribution by treatment and rounds

Widgets in P1	Round 1-5	Round 6-10	Total	Chi-2 Test	Mann-Whitney
Supply Shock					
0	213	183	396	4.6129	-1.5061
1	105	127	232	Pr = 0.100	Pr = 0.106
2	122	130	252		
SS + Price Cap					
0	211	180	391	4.7154	-1.6152
1	108	129	237	Pr = 0.095	Pr = 0.083
2	121	131	252		
Liberalization					
0	262	232	494	6.1290	-1.4172
1	112	145	257	Pr = 0.047	Pr = 0.112
2	66	63	129		

Table 7: Shortage Rate by Period

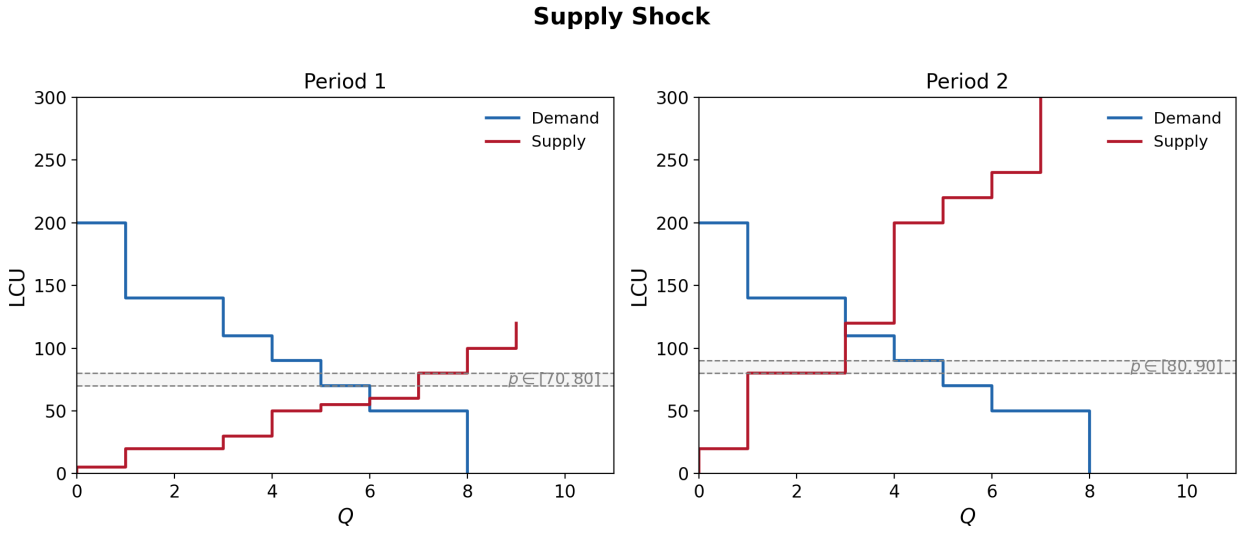
Treatment	Period	N	Shortage Rate	se	95% CI
Supply Shock	1	70	0.04	0.008	[0.02, 0.05]
	2	70	0.03	0.012	[0.01, 0.05]
SS + Price Cap	1	110	0.08	0.010	[0.06, 0.10]
	2	110	0.60	0.020	[0.56, 0.64]
Liberalization	1	110	0.25	0.017	[0.21, 0.28]
	2	110	0.02	0.008	[0.00, 0.04]

Note: Uses all 10 rounds. Shortage rate is the share of unsatisfied demand among buyers who bid at or above the transaction price. Period 1 bid prices are unavailable for 4 Supply Shock groups from early sessions (the oTree application recorded only the period 2 bid price). Period 2 bids are unavailable for buyers who stored (bought 2 units in period 1).

Table 8: Total Surplus and Efficiency Ratio

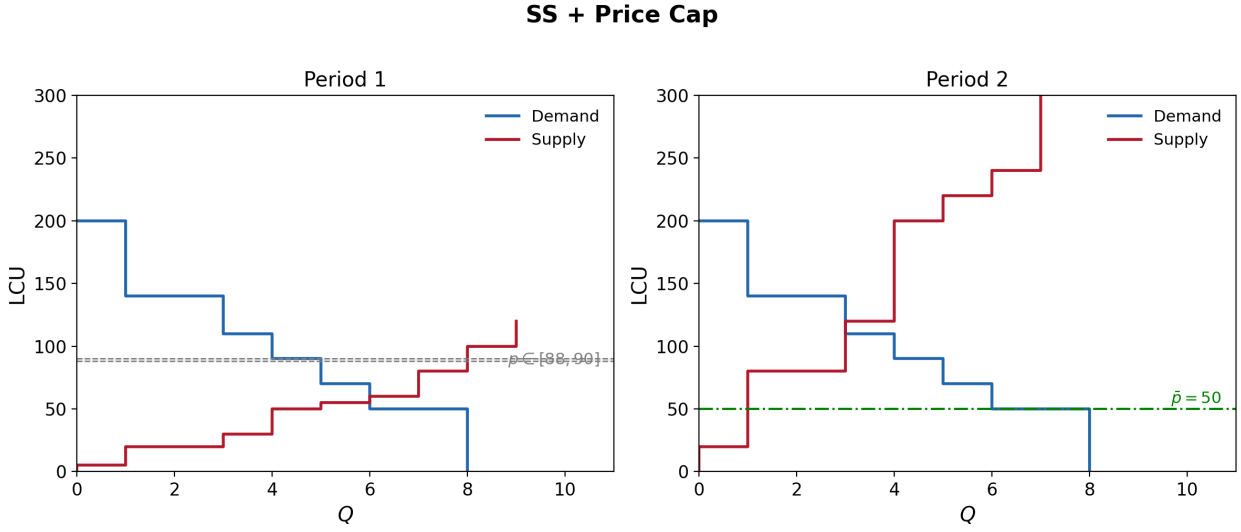
Treatment	N	Total Surplus	W^*	Efficiency (κ)	95% CI
Supply Shock	55	700.7	915.0	0.77	[0.73, 0.80]
SS + Price Cap	55	651.5	864.0	0.75	[0.72, 0.79]
Liberalization	55	609.2	793.1	0.77	[0.73, 0.81]

Note: Total surplus $W = \sum_i r_i q_i - \sum_j c_j y_j - \alpha s$, where $r_i q_i$ is the consumption value of buyer i, $c_j y_j$ is the production cost of seller j, α is the number of stored units, and $s = 10$ is the storage cost. W^* is the theoretical equilibrium total surplus. Uses data from last 5 rounds.



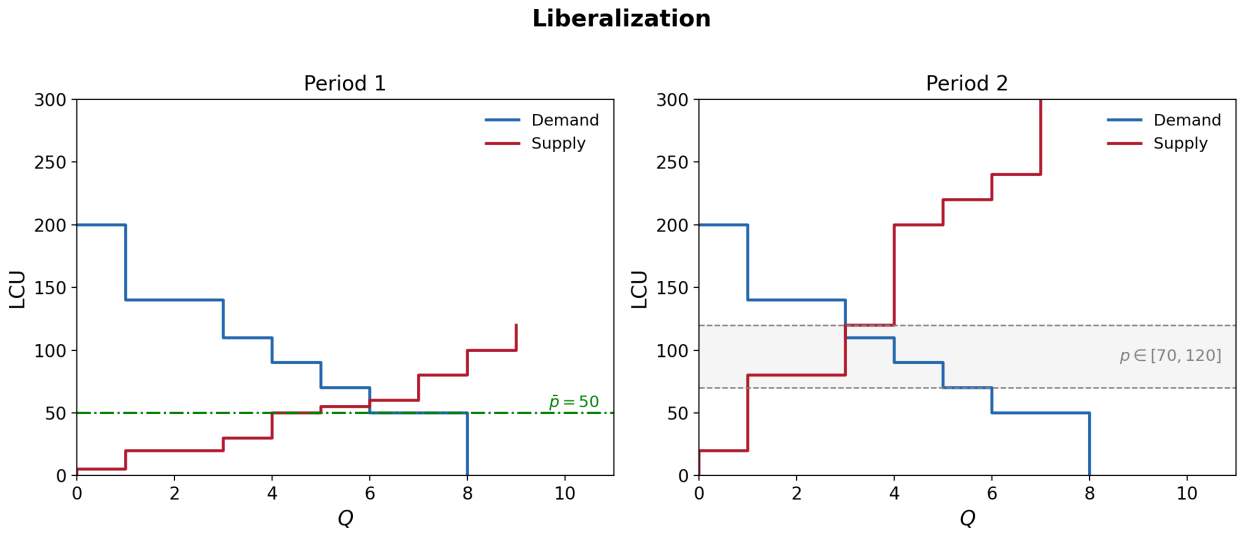
Notes: The figure displays the supply and demand schedules for both periods. In period 1, sellers' costs are $c_1(i)$ and the market operates freely. In period 2, costs increase fourfold to $c_2(i) = 4 \times c_1(i)$. The dashed horizontal lines indicate the competitive equilibrium price range.

Figure 2: Supply Shock



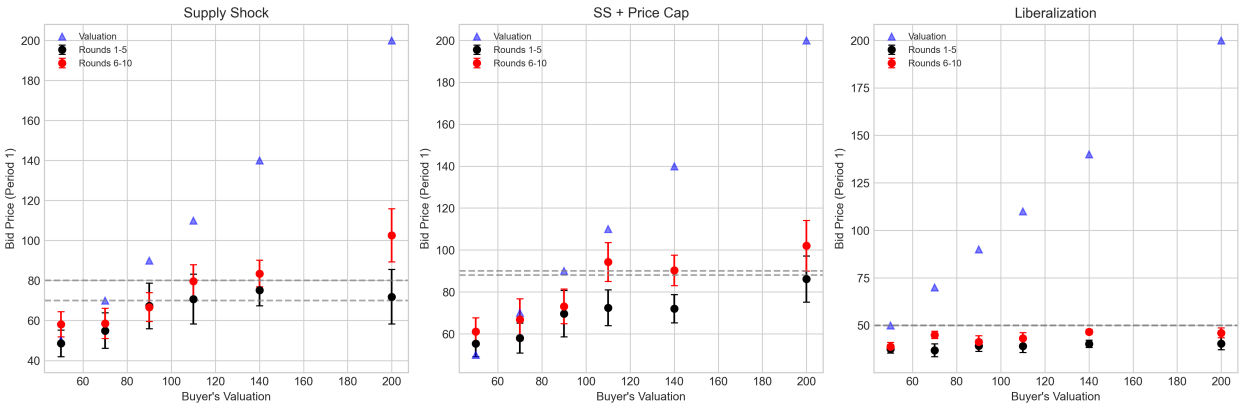
Notes: The figure displays the supply and demand schedules for both periods. In period 1, sellers' costs are $c_1(i)$ and the market operates freely; the dashed horizontal lines indicate the rationing equilibrium price range $p_1^* \in [88, 90]$. In period 2, costs increase fourfold to $c_2(i) = 4 \times c_1(i)$ and a price ceiling $\bar{p}_2 = 50$ is imposed (dash-dotted line). Only one seller has costs below the ceiling in period 2.

Figure 3: SS + Price Cap



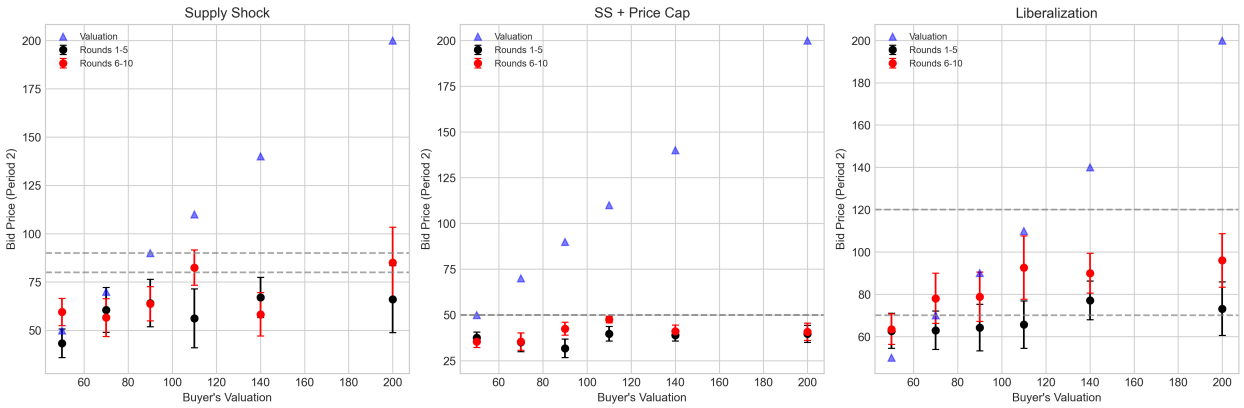
Notes: The figure displays the supply and demand schedules for both periods. In period 1, sellers' costs are $c_1(i)$ and a price ceiling $\bar{p}_1 = 50$ is imposed (dash-dotted line), creating a shortage. In period 2, costs increase fourfold to $c_2(i) = 4 \times c_1(i)$ and the market is liberalized; the dashed horizontal lines indicate the equilibrium price range $p_2^* \in [70, 120]$.

Figure 4: Liberalization



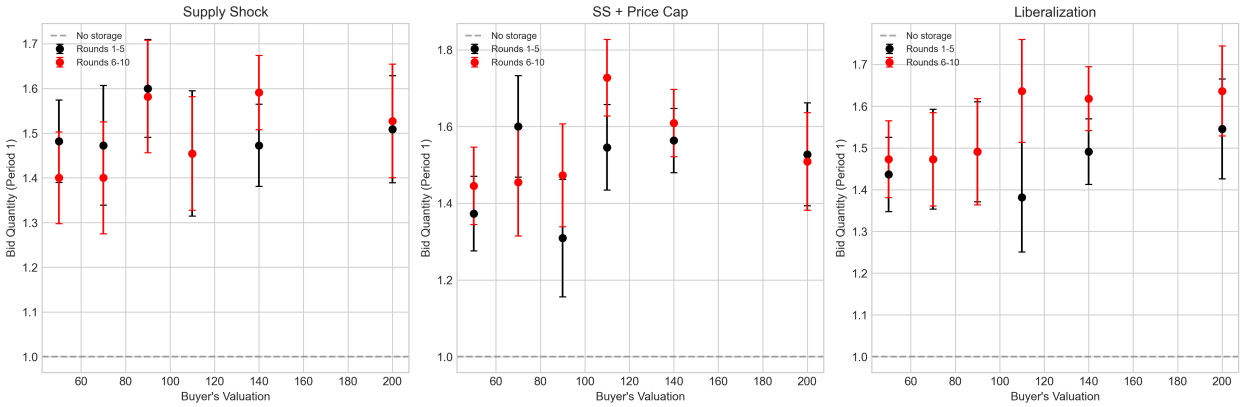
Notes: the figure presents the average bid price in all treatments. The lines represents the confidence intervals at a significance level of 90%. The dashed horizontal lines denote the range of equilibrium prices in each case.

Figure 5: Bid prices in period 1



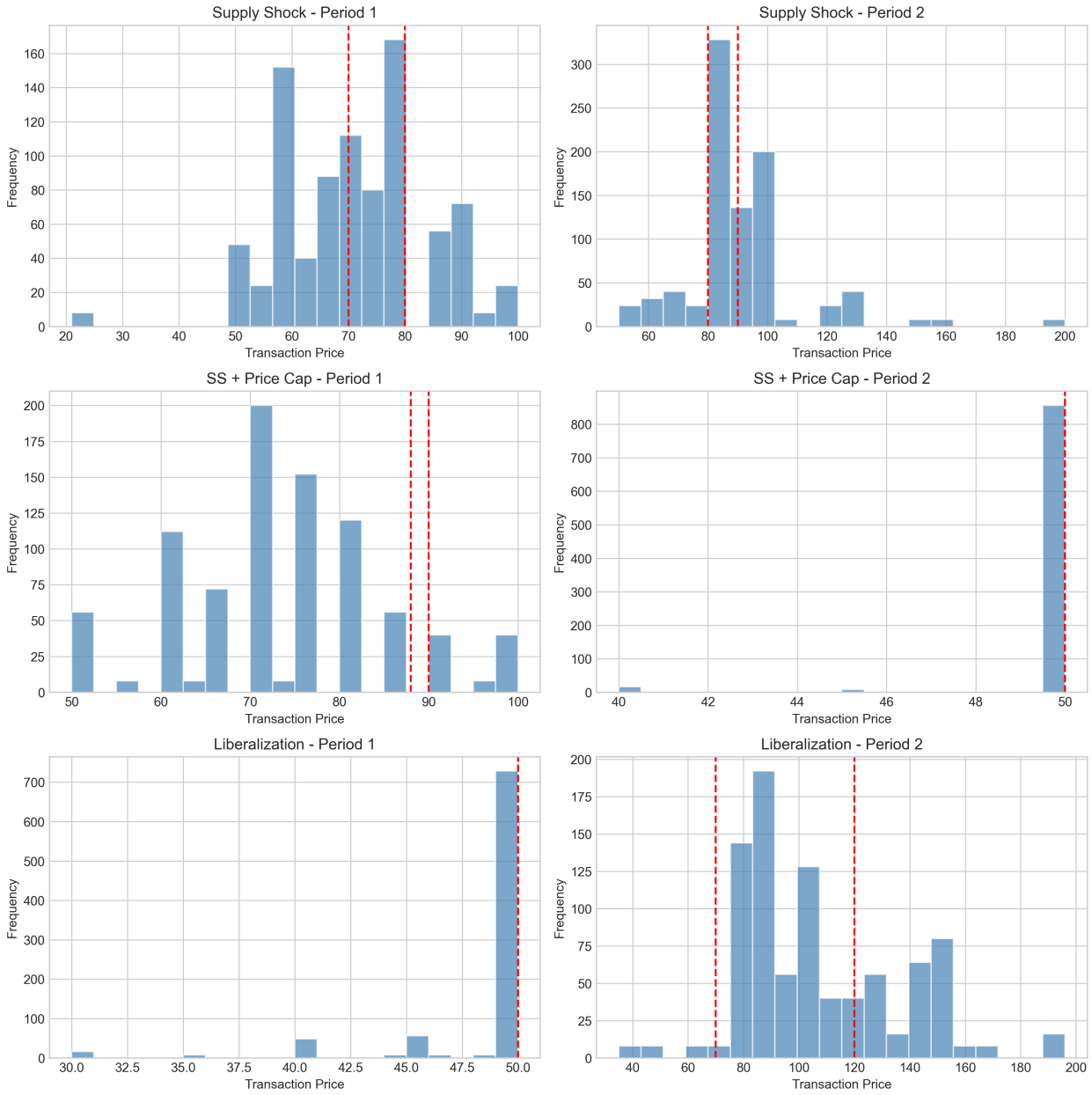
Notes: the figure presents the average bid price in all treatments. The vertical lines represents the confidence intervals at a significance level of 90%. The dashed horizontal lines denote the range of equilibrium prices in each case.

Figure 6: Bid prices in period 2



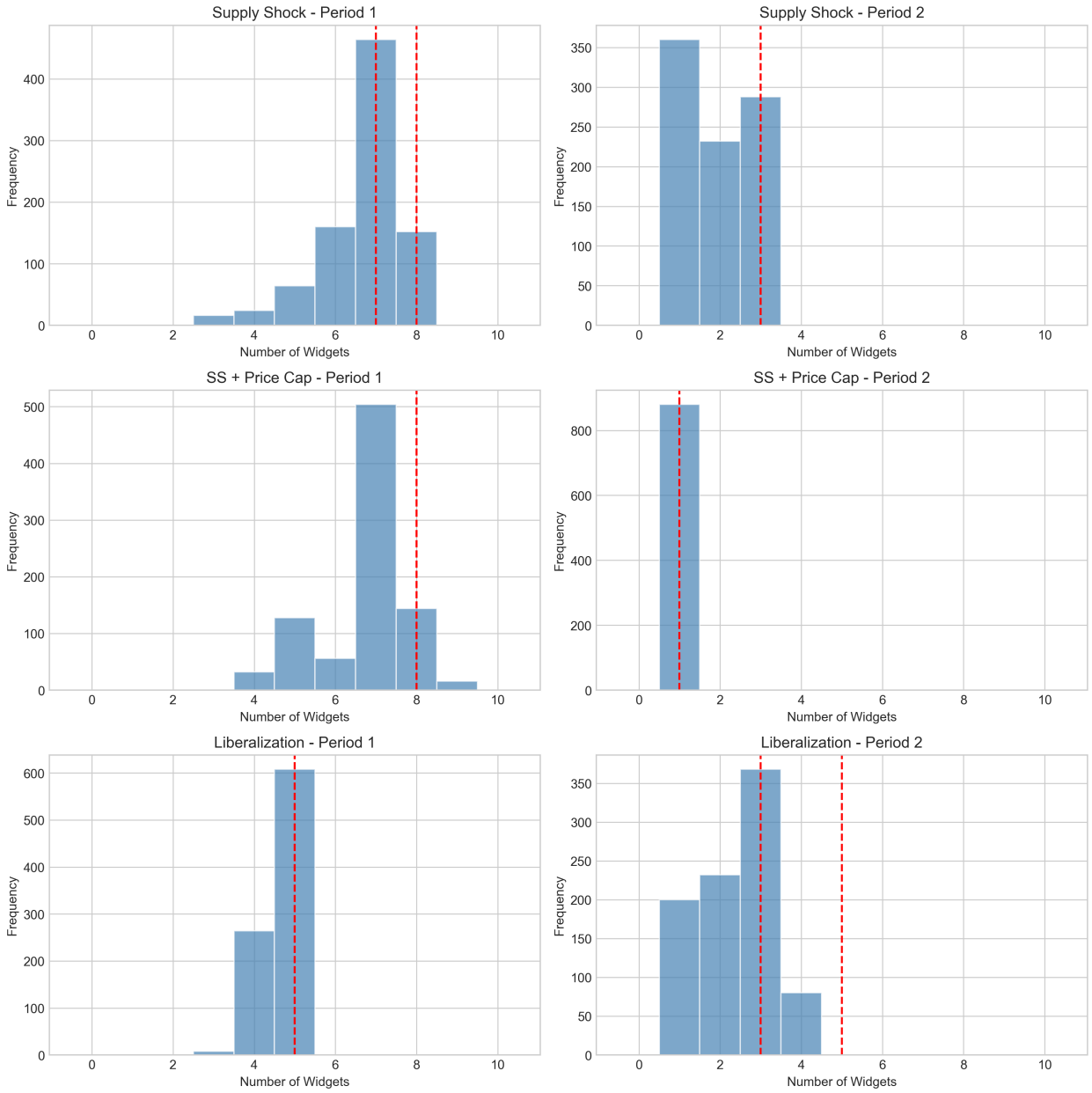
Notes: the figure presents the average quantity bid for all treatments in period 1. The vertical lines represents the confidence intervals at a significance level of 90%. The dashed horizontal line is set at 1 which means that the participant does not attempt to store goods.

Figure 7: Bid quantities in period 1



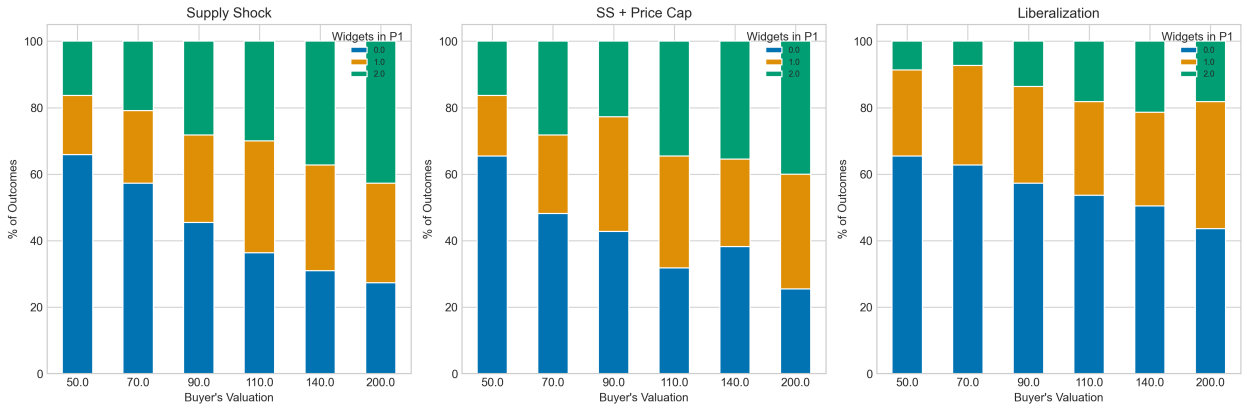
Note: The dashed vertical lines represent the competitive equilibrium for the supply shock treatment. For the price cap treatments, they indicate the rationing equilibrium.

Figure 8: Trading prices, last 5 rounds



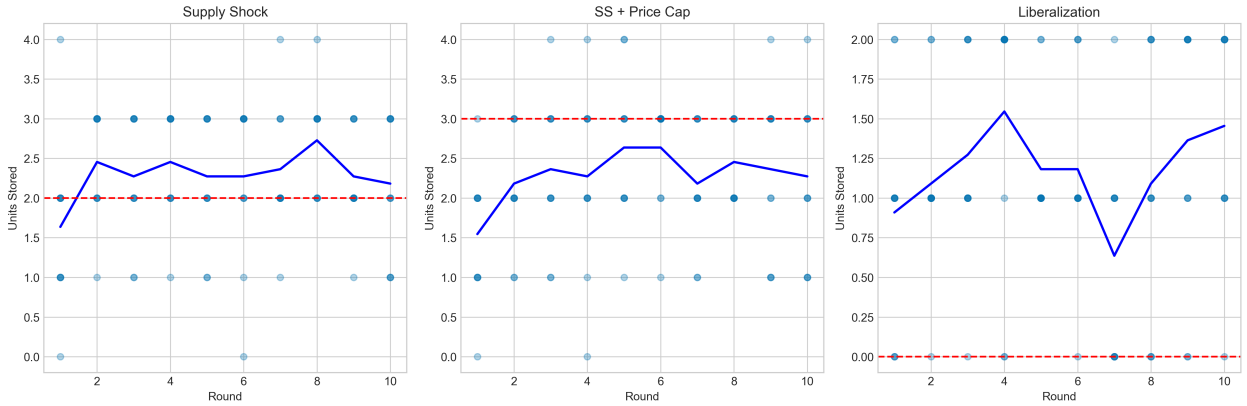
Note: The dashed vertical lines represent the competitive equilibrium for the supply shock treatment. For the price cap treatments, they indicate the rationing equilibrium.

Figure 9: Traded quantities, last 5 rounds



Notes: The figure presents the distribution of players based on the number of widgets acquired in period 1 across all rounds, categorized according to their valuation.

Figure 10: Acquired widgets in period 1 by player's valuation and treatment



Notes: The figure shows the number of widgets stored in each round for the three treatments in the experiment. The dashed line represents the theoretical prediction for each market.

Figure 11: Storage Market in all Treatments

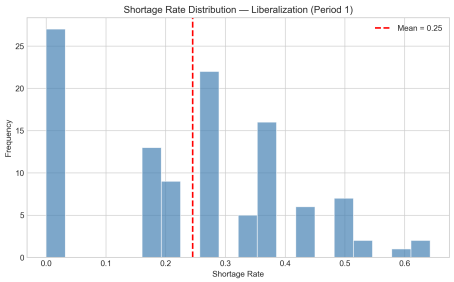
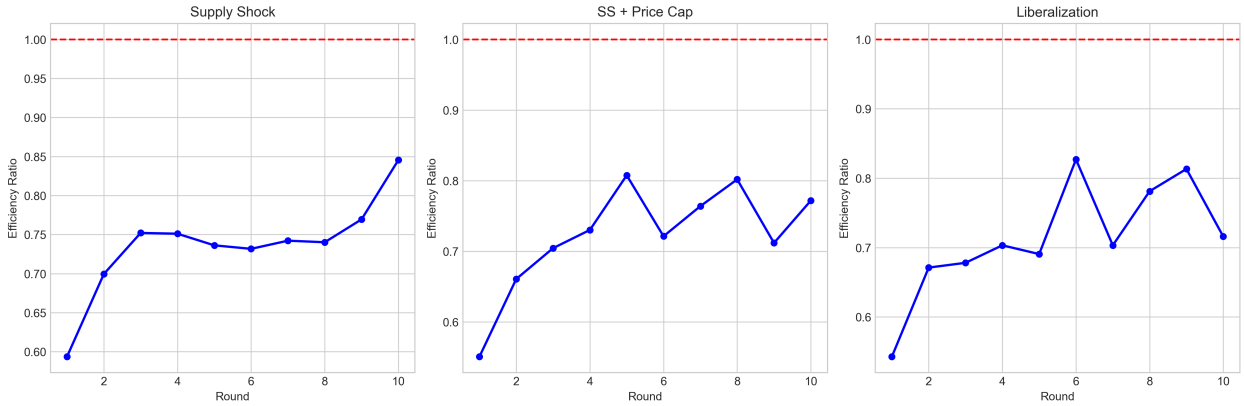


Figure 12: Shortage rate in the liberalization treatment (period 1)



Notes: The graphs represent the ratio of realized total surplus to theoretical total surplus for each treatment across rounds. The dashed line at 1.0 marks the theoretical benchmark.

Figure 13: Total surplus efficiency ratio over rounds

A Additional Tables and Figures

Table 9: Widget's value distribution by treatment, bidding quantity and rounds

Player's valuation	Round 1-5	Round 6-10	Total	Pearson's Chi-2 Test
Supply Shock				
Bidding Quantity in P1 = 0				
50	5	10	15	
70	3	2	5	
90	0	2	2	6.0639
110	4	2	6	Pr = 0.300
140	5	2	7	
200	1	2	3	
Bidding Quantity in P1 = 1				
50	47	46	93	
70	23	29	52	
90	22	19	41	1.9549
110	22	26	48	Pr = 0.855
140	48	41	89	
200	25	22	47	
Bidding Quantity in P1 = 2				
50	58	54	112	
70	29	24	53	
90	33	34	67	1.5656
110	29	27	56	Pr = 0.905
140	57	67	124	
200	29	31	60	

Table 9: Widget's value distribution by treatment, bidding quantity and rounds (continued)

Player's valuation	Round 1-5	Round 6-10	Total	Pearson's Chi-2 Test
SS + Price Cap				
Bidding Quantity in P1 = 0				
50	8	9	17	
70	3	4	7	
90	7	3	10	2.6474
110	0	0	0	Pr = 0.618
140	2	4	6	
200	3	2	5	
Bidding Quantity in P1 = 1				
50	53	43	96	
70	16	22	38	
90	24	23	47	4.4760
110	25	15	40	Pr = 0.483
140	44	35	79	
200	20	23	43	
Bidding Quantity in P1 = 2				
50	49	58	107	
70	36	29	65	
90	24	29	53	2.8606
110	30	40	70	Pr = 0.721
140	64	71	135	
200	32	30	62	

Table 9: Widget's value distribution by treatment, bidding quantity and rounds (continued)

Player's valuation	Round 1-5	Round 6-10	Total	Pearson's Chi-2 Test
Liberalization				
Bidding Quantity in P1 = 0				
50	4	5	9	
70	1	0	1	
90	1	2	3	2.5990
110	3	2	5	Pr = 0.627
140	0	0	0	
200	1	0	1	
Bidding Quantity in P1 = 1				
50	54	48	102	
70	27	29	56	
90	26	24	50	2.8923
110	28	16	44	Pr = 0.717
140	56	42	98	
200	23	20	43	
Bidding Quantity in P1 = 2				
50	52	57	109	
70	27	26	53	
90	28	29	57	2.1286
110	24	37	61	Pr = 0.831
140	54	68	122	
200	31	35	66	

Notes: The table above provides an overview of the distribution of the value of the widgets during the first and last five rounds of the period for each of the evaluated treatments and for each quantity intended to acquire in period 1.

A.1 Robustness: All 10 Rounds

The main-text tables report results from the last 5 rounds (rounds 6–10) to allow for learning. The tables below replicate the key results using all 10 rounds. The qualitative conclusions are unchanged.

Table 10: Market Price

Treatment	Period	N	Price	se	95% CI	Equilibrium
Supply Shock	1	110	71.44	1.24	[69.01, 73.86]	[70, 80]
	2	110	90.28	1.99	[86.38, 94.18]	[80, 90]
SS + Price Cap	1	110	72.61	1.10	[70.46, 74.76]	[88, 90]
	2	110	49.77	0.14	[49.51, 50.04]	[50, 50]
Liberalization	1	110	48.51	0.37	[47.79, 49.23]	[50, 50]
	2	110	106.72	2.75	[101.33, 112.10]	[70, 120]

Note: Uses data from all 10 rounds

Table 11: Market Quantities

Treatment	Period	N	Q	se	95% CI	Equilibrium
Supply Shock	1	110	6.69	0.10	[6.50, 6.89]	[7, 8]
	2	110	1.92	0.08	[1.76, 2.08]	[3, 3]
SS + Price Cap	1	110	6.74	0.10	[6.54, 6.94]	[8, 8]
	2	110	1.00	0.00	[1.00, 1.00]	[1, 1]
Liberalization	1	110	4.68	0.05	[4.59, 4.77]	[5, 5]
	2	110	2.37	0.09	[2.20, 2.55]	[3, 5]

Note: Uses data from all 10 rounds

Table 12: Differences in market price and stored goods

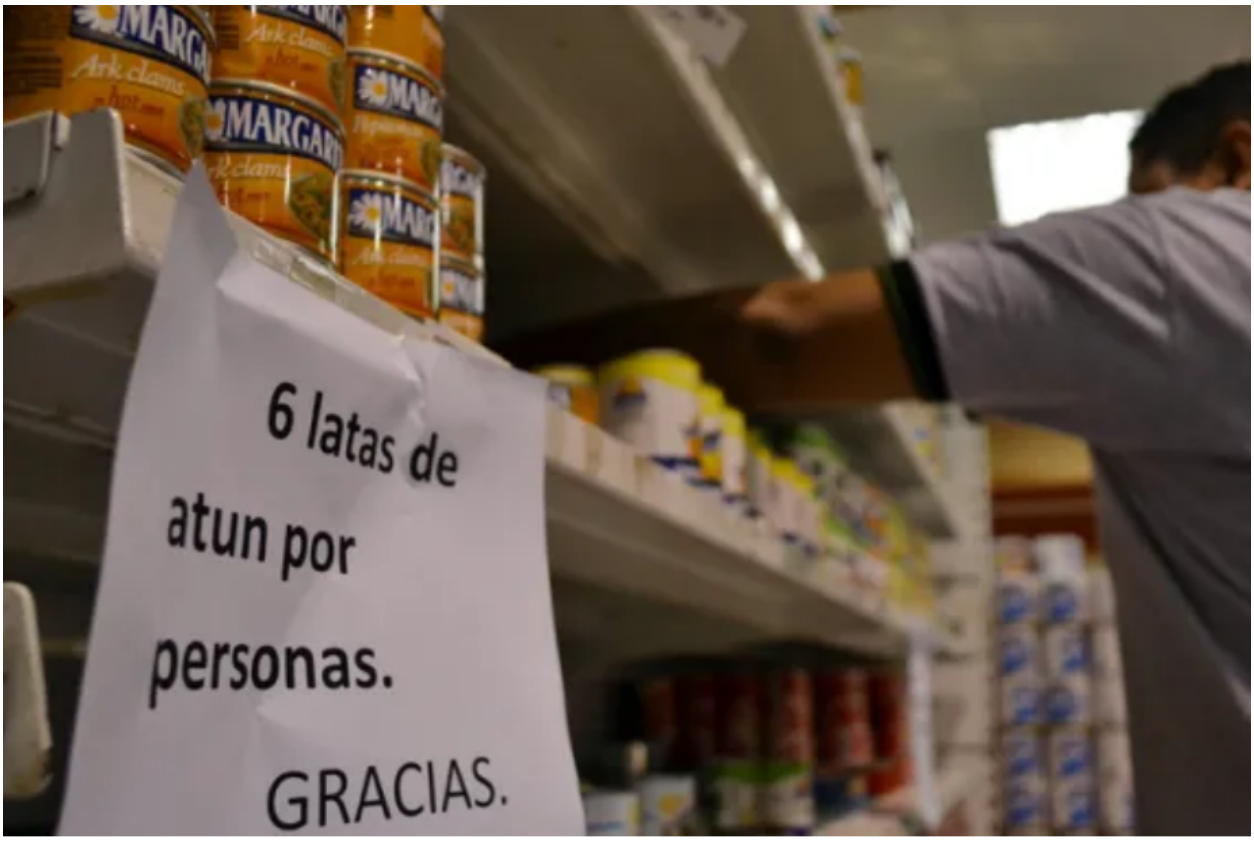
	Supply Shock	SS + Price Cap	Liberalization	T-Test	Mann-Whitney
Num. Markets	110	110	110	-	-
Price in $t = 1$	71.44 (1.235)	72.61 (1.099)	48.51 (0.367)	-0.7093 Pr = 0.4789	5758.5000 Pr = 0.5349
Stored Goods	2.29 (0.078)	2.29 (0.084)	1.17 (0.067)	0.0000 Pr = 1.0000	6065.0000 Pr = 0.9738

Note: Uses data from all 10 rounds. T-test and Mann-Whitney compare Supply Shock vs SS + Price Cap.

Table 13: Total Surplus and Efficiency Ratio

Treatment	N	Total Surplus	W^*	Efficiency (κ)	95% CI
Supply Shock	110	673.6	915.0	0.74	[0.71, 0.76]
SS + Price Cap	110	624.2	864.0	0.72	[0.69, 0.75]
Liberalization	110	565.2	793.1	0.71	[0.68, 0.75]

Note: Total surplus $W = \sum_i r_i q_i - \sum_j c_j y_j - \alpha s$, where $r_i q_i$ is the consumption value of buyer i, $c_j y_j$ is the production cost of seller j, α is the number of stored units, and $s = 10$ is the storage cost. W^* is the theoretical equilibrium total surplus. Uses data from all 10 rounds.



Note: Picture taken from Al Jazeera (2014), Retrieved on October 10, 2023.

Figure 14: No more than 6 cans of tuna per person, thank you

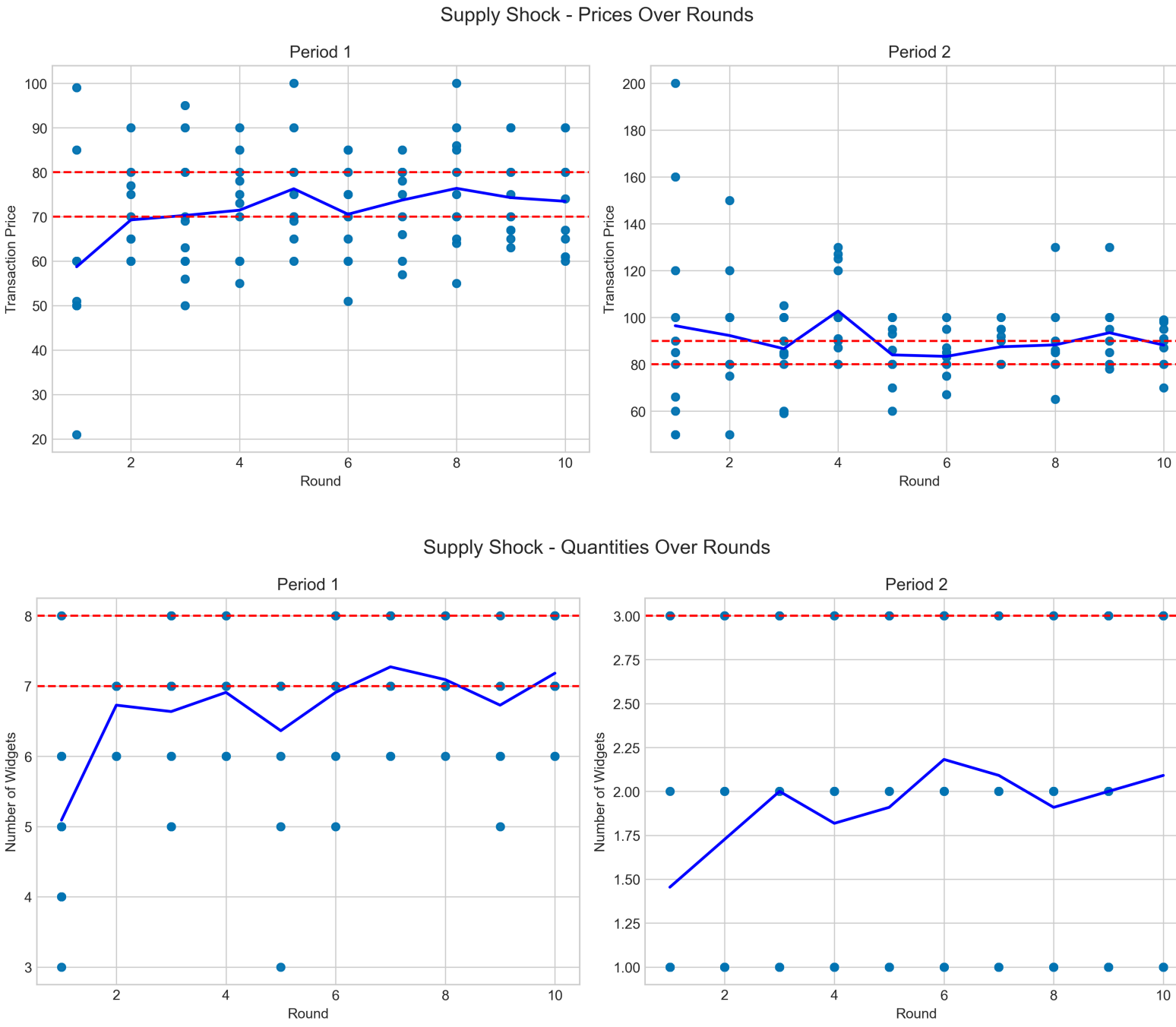


Figure 15: Supply Shock Market Outcomes

Notes: The figure illustrates the market outcomes for the indicated treatment. The first row displays the market price in each period, while the second row shows the quantities. The dashed horizontal line represents the theoretical prediction for each dimension. In cases where two lines are present, they indicate the upper and lower limits of the equilibrium values, respectively.

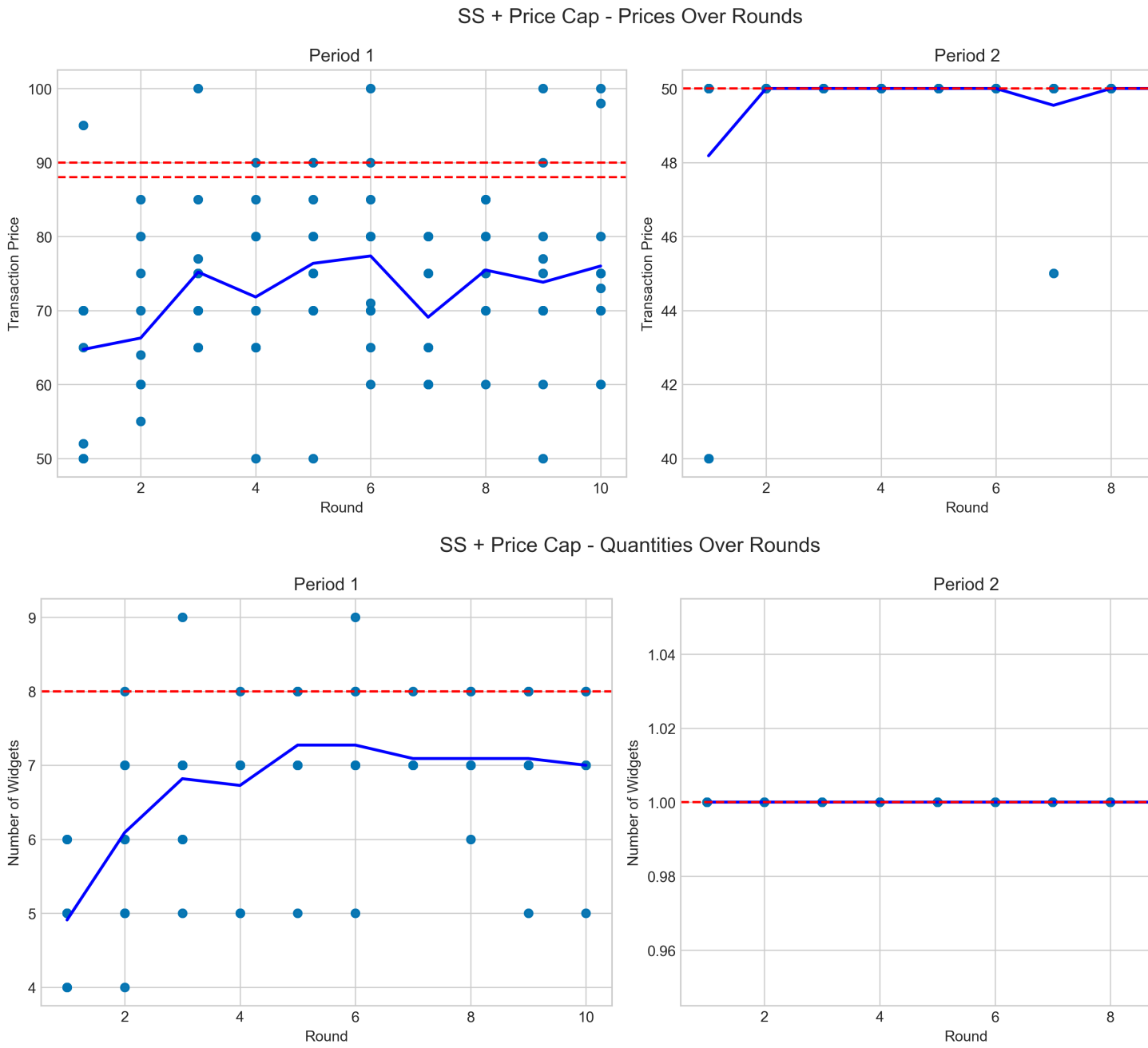


Figure 16: SS + Price Cap Treatment Outcomes

Notes: The figure illustrates the market outcomes for the indicated treatment. The first row displays the market price in each period, while the second row shows the quantities. The dashed horizontal line represents the theoretical prediction for each dimension. In cases where two lines are present, they indicate the upper and lower limits of the equilibrium values, respectively.

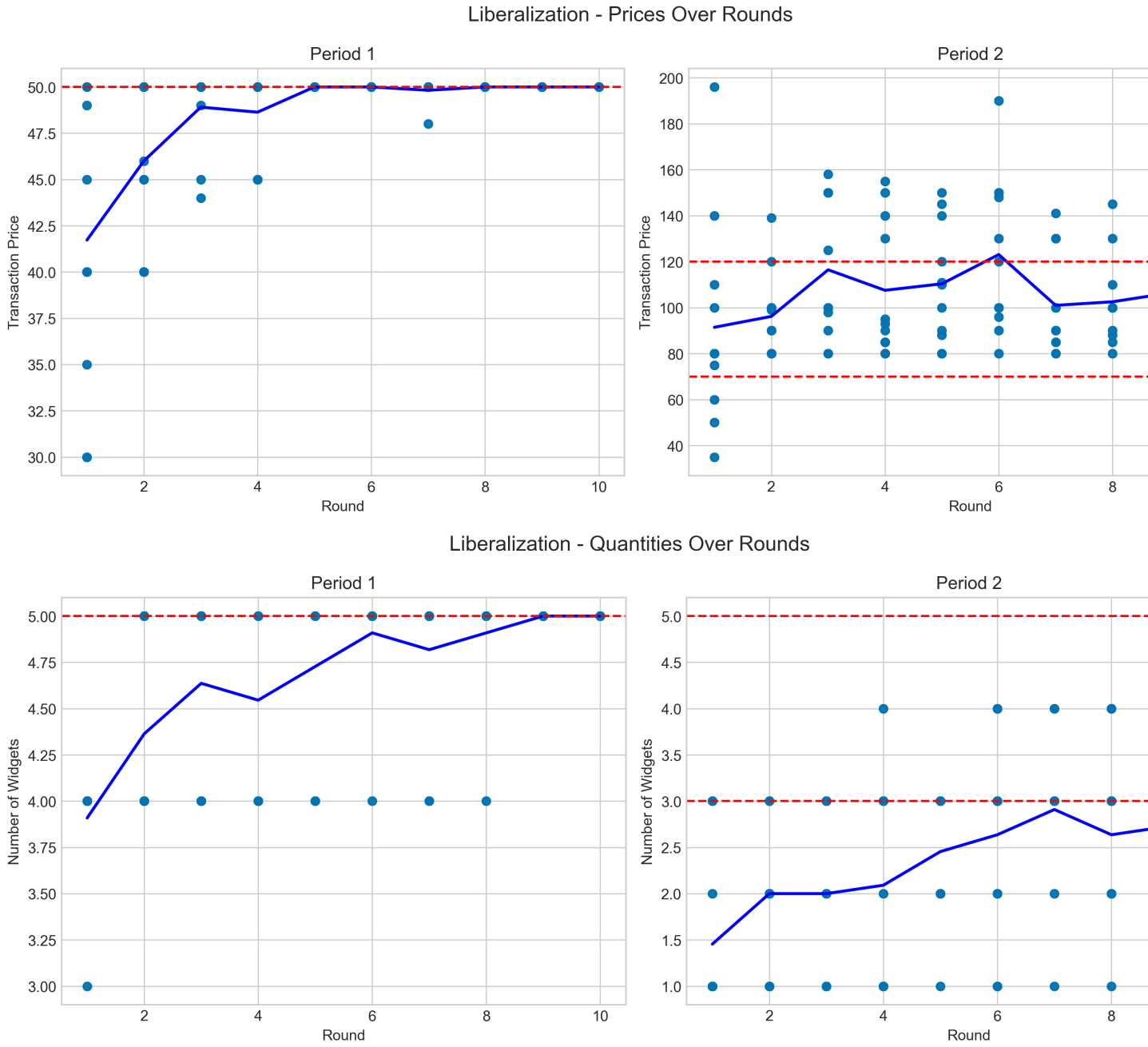


Figure 17: Liberalization Treatment Outcomes

Notes: The figure illustrates the market outcomes for the indicated treatment. The first row displays the market price in each period, while the second row shows the quantities. The dashed horizontal line represents the theoretical prediction for each dimension. In the cases where two lines are present, they indicate the upper and lower limits of the equilibrium values, respectively.

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